

Clustering Model Training Notebook

Setup Environment

```
In [289... # DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

Drive already mounted at /content/gdrive; to attempt to forcibly remount, call drive.mount("/content/gdrive", force_remount=True).

You can now save your data files in: /content/gdrive/MyDrive/36106/assignment/AT3/data

Student Information

```
In [290... group_name = "20"
student_name = "Shanu Sharma"
student_id = "25979360"
```

```
In [291... # Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

group_name

20

```
In [292... # Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)
```

student_name

Shanu Sharma

```
In [293... # Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)
```

student_id

25979360

0. Python Packages

0.a Install Additional Packages

If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
In [294... # No additional packages are required.
```

0.b Import Packages

```
In [295... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

B. Business Understanding

```
In [296... business_use_case_description = """
Segment 254 sales-active products into groups with similar commercial pro
the business can identify where margin risk is concentrated and prioritis
and portfolio interventions accordingly.
```

Each product is described across four commercial dimensions: structural a (product line, class, style, manufacture type), cost and price history (s list price, cost and price volatility), sales behaviour (total quantity s discount exposure (number of real discount offers, maximum discount rate, discount flag). Estimated margin percentage is derived from cost and pric direct measure of per-unit margin efficiency.

A product carrying high discount exposure alongside a thin margin is one pricing decision away from sustained margin erosion. Clustering makes tha visible by grouping products that share the same commercial pattern – bef

```
erosion shows up in revenue reporting.
"""
```

```
In [297... # Do not modify this code
print_tile(size="h3", key='business_use_case_description', value=business
business_use_case_description
```

Segment 254 sales-active products into groups with similar commercial profiles so the business can identify where margin risk is concentrated and prioritise pricing and portfolio interventions accordingly. Each product is described across four commercial dimensions: structural attributes (product line, class, style, manufacture type), cost and price history (standard cost, list price, cost and price volatility), sales behaviour (total quantity sold), and discount exposure (number of real discount offers, maximum discount rate, volume discount flag). Estimated margin percentage is derived from cost and price as the direct measure of per-unit margin efficiency. A product carrying high discount exposure alongside a thin margin is one missed pricing decision away from sustained margin erosion. Clustering makes that risk visible by grouping products that share the same commercial pattern — before the erosion shows up in revenue reporting.

```
In [298... business_objectives = """
The primary objective is to produce interpretable product segments that a
managers and pricing teams to prioritise margin interventions across the

Accurate clustering enables the business to:
- Identify products with high discount exposure and thin margins before t
a revenue problem
- Distinguish structural volume discounts (deliberate pricing strategy) f
discounts (margin erosion under demand pressure)
- Understand which product lines and cost tiers cluster together, informi
rationalisation decisions

Inaccurate clustering – where products with fundamentally different risk
grouped together – leads to undifferentiated pricing decisions and missed
to protect margin on at-risk products. For a portfolio where list prices
$2.29 to $3,578 (a 1,500× range), treating a high-value product the same
low-value one in pricing decisions carries disproportionate financial exp
"""
```

```
In [299... # Do not modify this code
print_tile(size="h3", key='business_objectives', value=business_objective
```

The primary objective is to produce interpretable product segments that allow category managers and pricing teams to prioritise margin interventions across the portfolio. Accurate clustering enables the business to:

- Identify products with high discount exposure and thin margins before they become a revenue problem
- Distinguish structural volume discounts (deliberate pricing strategy) from clearance discounts (margin erosion under demand pressure)
- Understand which product lines and cost tiers cluster together, informing portfolio rationalisation decisions

Inaccurate clustering — where products with fundamentally different risk profiles are grouped together — leads to undifferentiated pricing decisions and missed opportunities to protect margin on at-risk products. For a portfolio where list prices span from 2.29 to 3,578 (a 1,500x range), treating a high-value product the same as a low-value one in pricing decisions carries disproportionate financial exposure.

```
In [300... stakeholders_expectations_explanations = """
The primary users of the clustering output are category managers and pricing teams.

Category managers will use cluster assignments to review portfolio structure,
understanding whether high-cost products command proportionally higher prices,
and which product lines are exposed to discount pressure. Assignments feed in
to portfolio planning sessions where decisions on product rationalisation and
reviews are made.

Pricing teams will use centroid profiles (average feature values per cluster) to
design differentiated pricing strategies. Clusters with high max_discount
estimated_margin_pct are candidates for discount guardrails; clusters with
total_quantity_sold and healthy margins support volume-based pricing. The
drive the discount band and pricing tier decisions applied at the next period.

The results must be interpretable in business terms, not just mathematical.
A cluster solution that cannot be described in terms of product characteristics or
commercial behaviour has no actionable value for either stakeholder group.
"""

In [301... # Do not modify this code
print_tile(size="h3", key='stakeholders_expectations_explanations', value=stakeholders_expectations_explanations)
```

The primary users of the clustering output are category managers and pricing teams. Category managers will use cluster assignments to review portfolio structure — understanding whether high-cost products command proportionally higher prices, and which product lines are exposed to discount pressure. Assignments feed into quarterly portfolio planning sessions where decisions on product rationalisation and range reviews are made. Pricing teams will use centroid profiles (average feature values per cluster) to design differentiated pricing strategies. Clusters with high `max_discount_pct` and low `estimated_margin_pct` are candidates for discount guardrails; clusters with strong `total_quantity_sold` and healthy margins support volume-based pricing. These profiles drive the discount band and pricing tier decisions applied at the next pricing cycle. The results must be interpretable in business terms, not just mathematically valid. A cluster solution that cannot be described in terms of product characteristics and commercial behaviour has no actionable value for either stakeholder group.

```
In [302... business_hypothesis = """
          Meaningful margin-risk archetypes exist within the product portfolio and
          """

In [303... # Do not modify this code
          print_tile(size="h3", key="business_hypothesis", value=business_hypothesis)

          business_hypothesis
```

Meaningful margin-risk archetypes exist within the product portfolio and can be identified from commercial feature data alone through unsupervised clustering, producing between three and six commercially distinct segments with a Silhouette coefficient above 0.15.

C. Data Understanding

C.1 Load Datasets

```
In [304... X_train = pd.read_csv(at.folder_path / 'X_train.csv')
          print(f'X_train loaded: {X_train.shape[0]} products × {X_train.shape[1]} -
          X_train.head()

          X_train loaded: 254 products × 20 features (+ product_id)
```

	product_id	class	days_to_manufacture	is_manufactured	latest_standar
0	001a3a89-15cc-435e-89e5-0c844f44f595	-0.853875	-1.150517	0.0	-0
1	0025c2c9-855d-45a7-92ad-f4ab2fb016d1	0.299765	-0.542967	1.0	-0.
2	00d697bd-e064-4e0a-8486-c04f6a4377c3	-0.853875	-0.542967	1.0	-0.
3	01225971-c68c-4650-8d3c-f498ece3c5da	1.453405	1.279681	1.0	3
4	012fbb46-4cd3-4288-b1fa-d6fe3203db0f	-0.853875	-1.150517	0.0	-0.

5 rows × 21 columns

C.2 Define Target variable

```
In [305... # Clustering is an unsupervised task – no target variable exists.
# target_definition_explanations below explains this decision.
```

In [305...

```
In [306... target_definition_explanations = """
Clustering is an unsupervised learning task: no labelled target variable
and none is created. The objective is to discover natural groupings in th
feature space without reference to a pre-defined outcome.

In a supervised context, a target variable encodes the outcome the model
predict. Here, the 'outcome' – cluster membership – is itself what the mo
Defining a target before fitting would defeat the purpose of exploratory
and introduce label leakage into what should be a discovery exercise.

The placeholder target_name = 'y' is retained below for structural consis
notebook template; it is not used in any modelling step.
"""
```

```
In [307... # Do not modify this code
print_tile(size="h3", key='target_definition_explanations', value=target_
```

Clustering is an unsupervised learning task: no labelled target variable exists and none is created. The objective is to discover natural groupings in the product feature space without reference to a pre-defined outcome. In a supervised context, a target variable encodes the outcome the model is asked to predict. Here, the 'outcome' — cluster membership — is itself what the model produces. Defining a target before fitting would defeat the purpose of exploratory segmentation and introduce label leakage into what should be a discovery exercise. The placeholder `target_name = 'y'` is retained below for structural consistency with the notebook template; it is not used in any modelling step.

C.3 Create Target variable

```
In [308... # Not use for clustering.
#target_name = 'y'
```

C.4 Explore Target variable

```
In [309... # No target variable exists – target distribution exploration is not appl
```

```
In [309...
```

```
In [310... target_distribution_explanations = """
Target distribution analysis is not applicable to this unsupervised task.

In a supervised setting, examining the target distribution reveals class
value ranges, and skew – all of which affect model selection and evaluati
clustering, there is no target distribution to inspect. Cluster labels ar
by the algorithm after fitting and are evaluated using internal validity
(silhouette coefficient, Calinski-Harabász index, Davies-Bouldin index) r
comparison against a ground truth.
"""
```

```
In [311... # Do not modify this code
print_tile(size="h3", key='target_distribution_explanations', value=targe
```

target_distribution_explanations

Target distribution analysis is not applicable to this unsupervised task. In a supervised setting, examining the target distribution reveals class imbalance, value ranges, and skew — all of which affect model selection and evaluation. For clustering, there is no target distribution to inspect. Cluster labels are produced by the algorithm after fitting and are evaluated using internal validity indices (silhouette coefficient, Calinski-Harabász index, Davies-Bouldin index) rather than comparison against a ground truth.

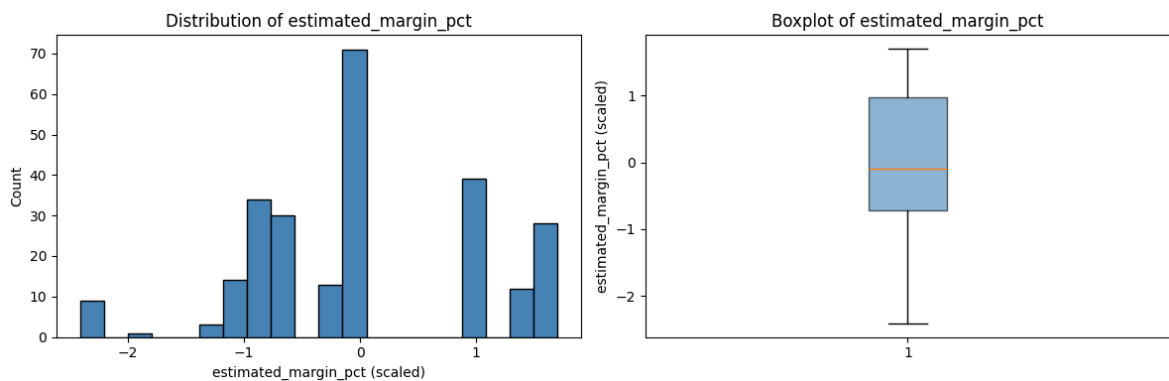
C.5 Explore Feature of Interest `estimated_margin_pct`

```
In [312... fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].hist(X_train['estimated_margin_pct'], bins=20, edgecolor='black',
axes[0].set_xlabel('estimated_margin_pct (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of estimated_margin_pct')

axes[1].boxplot(X_train['estimated_margin_pct'].dropna(), vert=True, patch_artist=True,
                boxprops=dict(facecolor='steelblue', alpha=0.6))
axes[1].set_ylabel('estimated_margin_pct (scaled)')
axes[1].set_title('Boxplot of estimated_margin_pct')

plt.tight_layout()
plt.show()
print(X_train['estimated_margin_pct'].describe().round(3))
```



```
count      254.000
mean         0.000
std          1.002
min         -2.410
25%         -0.717
50%         -0.102
75%          0.972
max          1.699
Name: estimated_margin_pct, dtype: float64
```

In [312...

```
In [313... feature_1_insights = """
estimated_margin_pct measures how much of the list price is retained as g
after subtracting standard production cost. It is the most direct indicat
profitability risk in this dataset.
```

Pre-scaling distribution (254 products): range 23%–63%, mean 46%, median 30%. The distribution is approximately symmetric – no strong skew – meaning Stata preserves the shape well and the scaled version is centred at zero with u

12 products sit below the 30% margin threshold, a level that typically si gnals pressure or below-market pricing. At the other end, 117 products exceed 40% margin indicating healthy pricing power or low production costs. No product has a negative margin in the training set, so there are no cost-exceeds-price cases to f

The 10-percentage-point standard deviation is wide enough to support mean separation: a cluster centred around 30–35% margin has materially differe nt characteristics than one centred around 55–60%. Distinguishing these grou


```
objective of the use case.
"""
```

```
In [314... # Do not modify this code
print_tile(size="h3", key='feature_1_insights', value=feature_1_insights)
```

feature_1_insights

estimated_margin_pct measures how much of the list price is retained as gross margin after subtracting standard production cost. It is the most direct indicator of per-unit profitability risk in this dataset. Pre-scaling distribution (254 products): range 23%–63%, mean 46%, median 45%, std 10%. The distribution is approximately symmetric — no strong skew — meaning StandardScaler preserves the shape well and the scaled version is centred at zero with unit variance. 12 products sit below the 30% margin threshold, a level that typically signals cost pressure or below-market pricing. At the other end, 117 products exceed 45% margin, indicating healthy pricing power or low production costs. No product has a negative margin in the training set, so there are no cost-exceeds-price cases to flag. The 10-percentage-point standard deviation is wide enough to support meaningful cluster separation: a cluster centred around 30–35% margin has materially different pricing-risk characteristics than one centred around 55–60%. Distinguishing these groups is a primary objective of the use case.

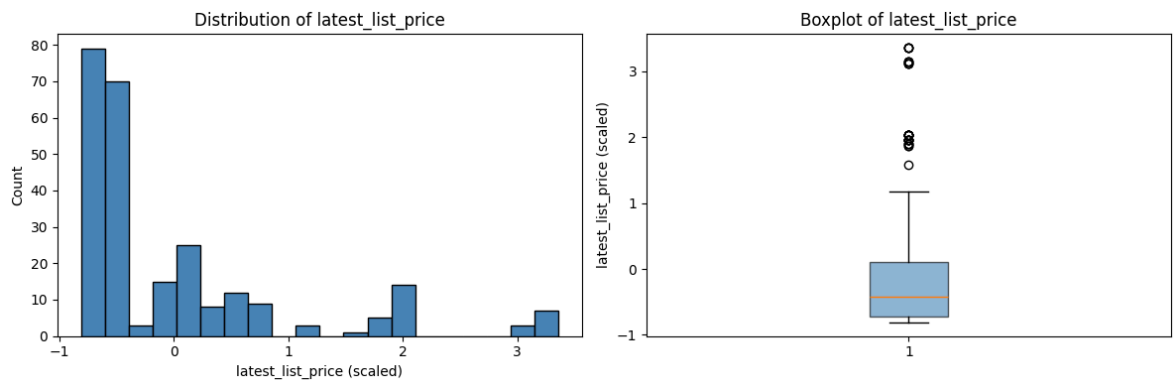
C.6 Explore Feature of Interest latest_list_price

```
In [315... fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].hist(X_train['latest_list_price'], bins=20, edgecolor='black', co
axes[0].set_xlabel('latest_list_price (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of latest_list_price')

axes[1].boxplot(X_train['latest_list_price'].dropna(), vert=True, patch_a
                boxprops=dict(facecolor='steelblue', alpha=0.6))
axes[1].set_ylabel('latest_list_price (scaled)')
axes[1].set_title('Boxplot of latest_list_price')

plt.tight_layout()
plt.show()
print(X_train['latest_list_price'].describe().round(3))
```



```
count    254.000
mean      0.000
std       1.002
min       -0.809
25%      -0.715
50%      -0.419
75%       0.101
max       3.358
Name: latest_list_price, dtype: float64
```

```
In [316... feature_2_insights = """
latest_list_price is the most recent catalogue price for each product. It
price tier a product occupies and, in combination with standard cost, dri
calculation directly.

Pre-scaling distribution (254 products): range \$2.29-\$3,578.27, median
std \$910. The distribution is heavily right-skewed – most products clust
price range while a minority of high-value products extend the tail. Stan
remove this skew; the scaled distribution retains the same shape.

Product tiers visible in the raw data: 68 products below \$100 (consumabl
91 between \$100-\$1,000 (mid-range), and 62 above \$1,000 (premium / hig
correspond closely to product line composition – S and T lines tend to co
price points, while R-line products span a wider range.

Price tier is a strong structural differentiator for clustering: high-pri
face different inventory risk, different discount pressure, and different
low-price products. Clusters that separate on price will therefore tend t
in business terms without additional labelling.
"""
```

```
In [317... # Do not modify this code
print_tile(size="h3", key='feature_2_insights', value=feature_2_insights)
```

latest_list_price is the most recent catalogue price for each product. It captures the price tier a product occupies and, in combination with standard cost, drives the margin calculation directly. Pre-scaling distribution (254 products): range \$2.29–\$3,578.27, median \$337, mean \$749, std \$910. The distribution is heavily right-skewed — most products cluster in the low-to-mid price range while a minority of high-value products extend the tail. StandardScaler does not remove this skew; the scaled distribution retains the same shape. Product tiers visible in the raw data: 68 products below \$100 (consumables and accessories), 91 between \$100–\$1,000 (mid-range), and 62 above \$1,000 (premium / high-end). These tiers correspond closely to product line composition — S and T lines tend to concentrate at lower price points, while R-line products span a wider range. Price tier is a strong structural differentiator for clustering: high-price products typically face different inventory risk, different discount pressure, and different margin dynamics than low-price products. Clusters that separate on price will therefore tend to be interpretable in business terms without additional labelling.

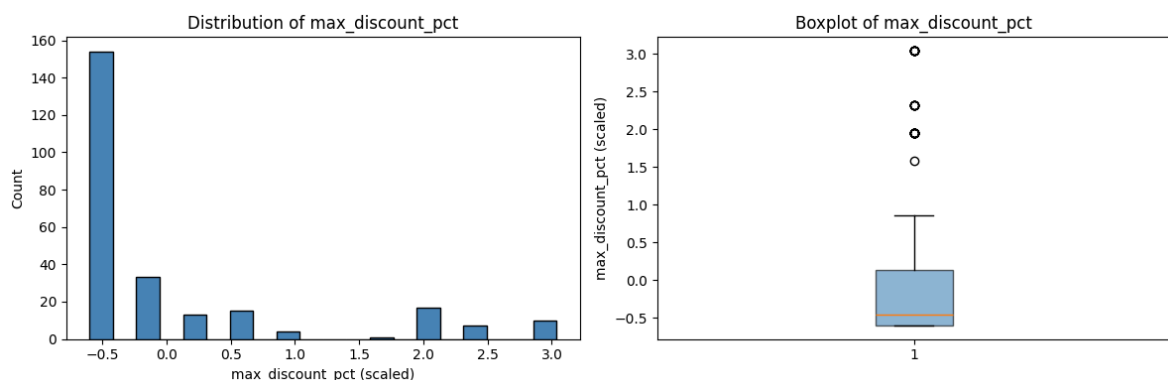
C.7 Explore Feature of Interest max_discount_pct

```
In [318]: fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].hist(X_train['max_discount_pct'], bins=20, edgecolor='black', col
axes[0].set_xlabel('max_discount_pct (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of max_discount_pct')

axes[1].boxplot(X_train['max_discount_pct'], vert=True, patch_artist=True
                boxprops=dict(facecolor='steelblue', alpha=0.6))
axes[1].set_ylabel('max_discount_pct (scaled)')
axes[1].set_title('Boxplot of max_discount_pct')

plt.tight_layout()
plt.show()
print(X_train['max_discount_pct'].describe().round(3))
print(f"\nProducts with no discount: {(X_train['max_discount_pct'] == X_t
```



```
count      254.000
mean       -0.000
std         1.002
min        -0.598
25%        -0.598
50%        -0.453
75%         0.129
max         3.039
Name: max_discount_pct, dtype: float64
```

Products with no discount: 112

In [318...

In [319...

```
feature_n_insights = """
max_discount_pct records the highest promotional discount rate ever offer
It is a proxy for demand support risk: products that require heavy discou
sales volume signal either weak pricing power, excess inventory, or both.

Pre-scaling distribution (254 products): 112 products have never been dis
(max_discount_pct = 0); 142 carry at least some discount history. Among t
distribution is right-skewed – most discounts are modest (10–20%), but 34
have faced discounts above 30%, and 10 products carry a 50% maximum disco
is the highest level in the retailer's promotion schedule and typically r
for clearance or excess-inventory disposal.

After scaling, the distribution becomes bimodal: a large spike at the low
(all zero-discount products mapped to the same z-score of approximately –
and a spread of values for products with real discount history. This bimo
structure is informative for clustering – the two groups have fundamental
commercial characteristics – but it also means the centroid of any cluste
this feature boundary will not represent either group well. Algorithms th
capture this discontinuity (such as hierarchical clustering) may handle i
than those that assume smooth, convex clusters.
"""
```

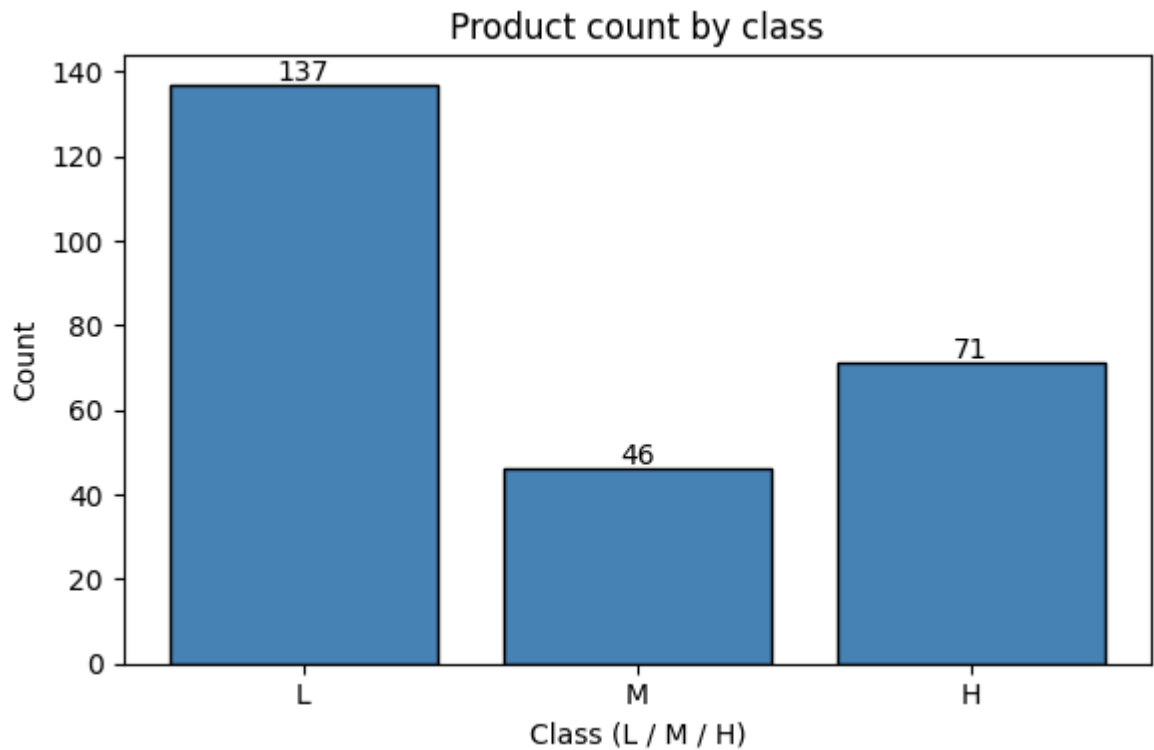
In [320...

```
# Do not modify this code
print_tile(size="h3", key='feature_n_insights', value=feature_n_insights)
```

`max_discount_pct` records the highest promotional discount rate ever offered on a product. It is a proxy for demand support risk: products that require heavy discounting to sustain sales volume signal either weak pricing power, excess inventory, or both. Pre-scaling distribution (254 products): 112 products have never been discounted (`max_discount_pct = 0`); 142 carry at least some discount history. Among those, the distribution is right-skewed — most discounts are modest (10–20%), but 34 products have faced discounts above 30%, and 10 products carry a 50% maximum discount, which is the highest level in the retailer's promotion schedule and typically reserved for clearance or excess-inventory disposal. After scaling, the distribution becomes bimodal: a large spike at the lower end (all zero-discount products mapped to the same z-score of approximately -0.60) and a spread of values for products with real discount history. This bimodal structure is informative for clustering — the two groups have fundamentally different commercial characteristics — but it also means the centroid of any cluster spanning this feature boundary will not represent either group well. Algorithms that can capture this discontinuity (such as hierarchical clustering) may handle it better than those that assume smooth, convex clusters.

C.8 Explore Feature of Interest `class`

```
In [321... class_labels = {X_train['class'].min(): 'L',
                  sorted(X_train['class'].unique())[1]: 'M',
                  X_train['class'].max(): 'H'}
counts = X_train['class'].value_counts().sort_index()
labels = [class_labels[v] for v in counts.index]
fig, ax = plt.subplots(figsize=(6, 4))
bars = ax.bar(labels, counts.values, edgecolor='black', color='steelblue')
for bar, v in zip(bars, counts.values):
    ax.text(bar.get_x() + bar.get_width()/2, v + 1, str(v), ha='center')
ax.set_xlabel('Class (L / M / H)')
ax.set_ylabel('Count')
ax.set_title('Product count by class')
plt.tight_layout()
plt.show()
print(dict(zip(labels, counts.values)))
```



```
{'L': np.int64(137), 'M': np.int64(46), 'H': np.int64(71)}
```

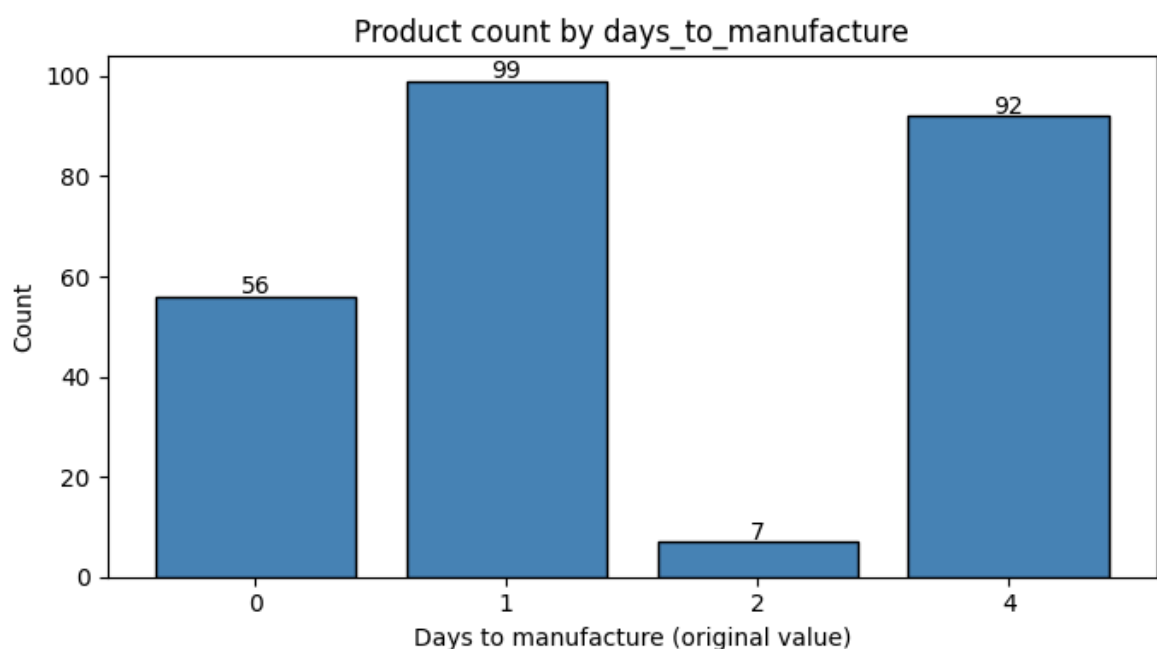
```
In [322... feature_3_insights = """  
class is an ordinal product quality tier encoded L=0, M=1, H=2 in SD—a an  
StandardScaled. The three scaled values map back to L (-0.854), M (0.300)  
  
Distribution: L=137 (54%), H=71 (28%), M=46 (18%). The majority of sales-  
products sit at the lowest quality tier. H-class products are the second-  
group despite being the highest complexity tier, reflecting that the port  
includes a substantial high-end segment alongside predominantly basic pro  
  
H-class products typically require more days to manufacture, carry higher  
costs, and command higher list prices, while L-class products tend to be  
high-volume items. The imbalanced distribution means class L will be the  
value in most clusters; H-class clusters, when they emerge, are commercia  
"""
```

```
In [323... # Do not modify this code  
print_tile(size='h3', key='feature_3_insights', value=feature_3_insights)
```

class is an ordinal product quality tier encoded L=0, M=1, H=2 in SD-a and then StandardScaled. The three scaled values map back to L (-0.854), M (0.300), H (1.453). Distribution: L=137 (54%), H=71 (28%), M=46 (18%). The majority of sales-active products sit at the lowest quality tier. H-class products are the second-largest group despite being the highest complexity tier, reflecting that the portfolio includes a substantial high-end segment alongside predominantly basic products. H-class products typically require more days to manufacture, carry higher standard costs, and command higher list prices, while L-class products tend to be low-cost, high-volume items. The imbalanced distribution means class L will be the modal value in most clusters; H-class clusters, when they emerge, are commercially distinct.

C.9 Explore Feature of Interest `days_to_manufacture`

```
In [324... # 4 distinct raw values: 0, 1, 2, 4 – plot as labelled bar chart
raw_vals = [0, 1, 2, 4]
raw_counts_dtm = [56, 99, 7, 92]
fig, ax = plt.subplots(figsize=(7, 4))
bars = ax.bar([str(v) for v in raw_vals], raw_counts_dtm, edgecolor='black')
for bar, v in zip(bars, raw_vals):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_xlabel('Days to manufacture (original value)')
ax.set_ylabel('Count')
ax.set_title('Product count by days_to_manufacture')
plt.tight_layout()
plt.show()
print(X_train['days_to_manufacture'].describe().round(3))
```



```

count      254.000
mean       -0.000
std         1.002
min        -1.151
25%        -0.543
50%        -0.543
75%         1.280
max         1.280
Name: days_to_manufacture, dtype: float64

```

```

In [325... feature_4_insights = """
days_to_manufacture records the production lead time for each product. It
proxy for manufacturing complexity and indirectly reflects cost structure

The feature takes only four distinct values: 0 (56 products – no producti
time, typically purchased items), 1 (99 products), 2 (7 products), and 4
The near-absence of 2-day products makes this effectively a three-tier va

Products requiring 4 days (92 products, 36%) tend to be structurally more
and concentrate in M-line and R-line. Zero-day products are predominantly
(is_manufactured=0) and carry lower standard costs. The feature was Stand
as a continuous variable because the differences between levels represent
production time differences rather than arbitrary category labels.
"""

```

```

In [326... # Do not modify this code
print_tile(size='h3', key='feature_4_insights', value=feature_4_insights)

```

feature_4_insights

days_to_manufacture records the production lead time for each product. It is a proxy for manufacturing complexity and indirectly reflects cost structure. The feature takes only four distinct values: 0 (56 products — no production lead time, typically purchased items), 1 (99 products), 2 (7 products), and 4 (92 products). The near-absence of 2-day products makes this effectively a three-tier variable. Products requiring 4 days (92 products, 36%) tend to be structurally more complex and concentrate in M-line and R-line. Zero-day products are predominantly purchased (is_manufactured=0) and carry lower standard costs. The feature was StandardScaled as a continuous variable because the differences between levels represent real production time differences rather than arbitrary category labels.

C.10 Explore Feature of Interest is_manufactured

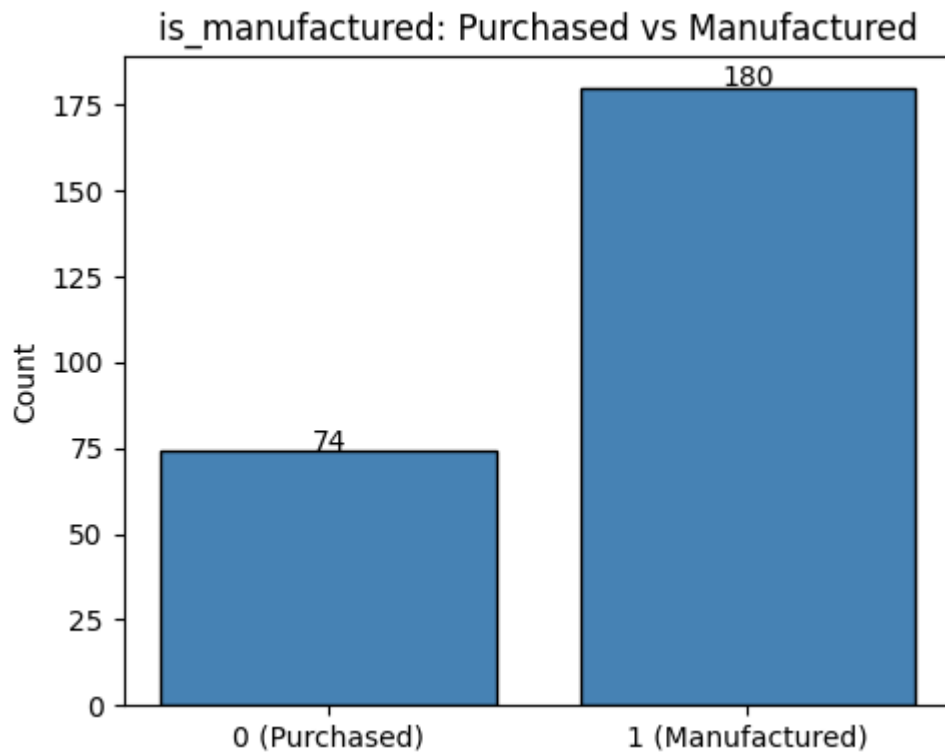
```

In [327... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['is_manufactured'].value_counts().sort_index()
bars = ax.bar(['0 (Purchased)', '1 (Manufactured)'], [counts.get(0,0), co
               edgecolor='black', color='steelblue'])
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('is_manufactured: Purchased vs Manufactured')

```



```
plt.tight_layout()
plt.show()
print(X_train['is_manufactured'].value_counts())
```



```
is_manufactured
1.0    180
0.0     74
Name: count, dtype: int64
```

In [328... `feature_5_insights = """`
`is_manufactured is a binary flag: 1 = produced in-house (180 products, 71`
`0 = externally purchased (74 products, 29%).`

Manufactured products carry production-driven cost structures – standard reflects materials and labour, and margin depends on production efficiency pricing decisions. Purchased products have externally-set costs with typically narrower cost variation and simpler margin dynamics.

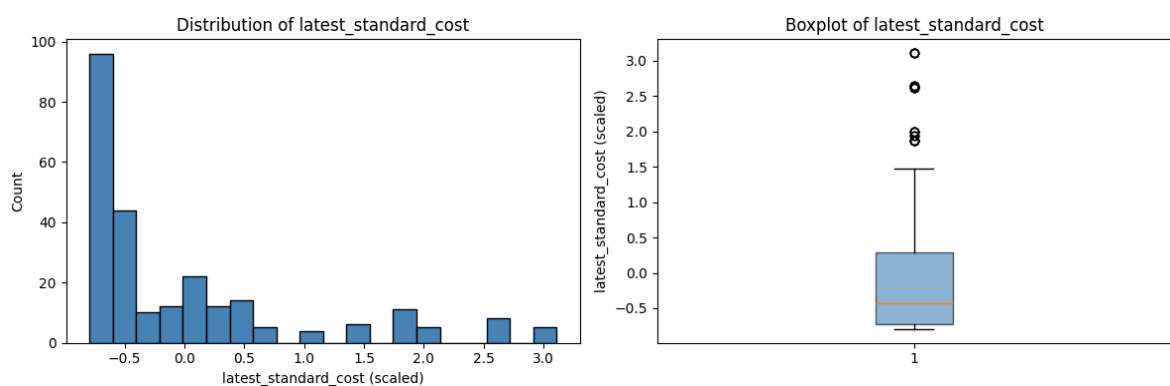
This distinction is relevant for margin risk interpretation: clusters that a mix of manufactured and purchased products have heterogeneous cost exposure. Manufactured products are exposed to internal cost changes; purchased products face supplier pricing risk. Identifying which cluster type a product belongs to informs whether the appropriate management lever is production efficiency or procurement negotiation.

In [329... `# Do not modify this code`
`print_tile(size='h3', key='feature_5_insights', value=feature_5_insights)`

is_manufactured is a binary flag: 1 = produced in-house (180 products, 71%), 0 = externally purchased (74 products, 29%). Manufactured products carry production-driven cost structures — standard cost reflects materials and labour, and margin depends on production efficiency and pricing decisions. Purchased products have externally-set costs with typically narrower cost variation and simpler margin dynamics. This distinction is relevant for margin risk interpretation: clusters that contain a mix of manufactured and purchased products have heterogeneous cost exposures. Manufactured products are exposed to internal cost changes; purchased products face supplier pricing risk. Identifying which cluster type a product belongs to informs whether the appropriate management lever is production efficiency or procurement negotiation.

C.11 Explore Feature of Interest latest_standard_cost

```
In [330... fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(X_train['latest_standard_cost'], bins=20, edgecolor='black',
axes[0].set_xlabel('latest_standard_cost (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of latest_standard_cost')
axes[1].boxplot(X_train['latest_standard_cost'].dropna(), vert=True, patc
                boxprops=dict(facecolor='steelblue', alpha=0.6))
axes[1].set_ylabel('latest_standard_cost (scaled)')
axes[1].set_title('Boxplot of latest_standard_cost')
plt.tight_layout()
plt.show()
print(X_train['latest_standard_cost'].describe().round(3))
```



```
count      254.000
mean         0.000
std         1.002
min        -0.793
25%        -0.730
50%        -0.436
75%         0.287
max         3.110
Name: latest_standard_cost, dtype: float64
```

```
In [331... feature_6_insights = """
latest_standard_cost is the most recent production cost. Together with
latest_list_price it is one of the two inputs to estimated_margin_pct.

Pre-scaling distribution: range \$0.86-\$2,171.29, median \$200, mean \$4
Heavily right-skewed – most products have relatively low production costs
minority of complex, high-value products carry very high costs.

latest_standard_cost is correlated with latest_list_price (r=0.90 in SD-a
was retained because cost structure adds information beyond price level:
with the same list price can have very different cost bases and therefore
different margin profiles. The correlation did not meet the 0.85 drop thr
the pair-wise check, and cost is directionally distinct from price in the
calculation.
"""
```

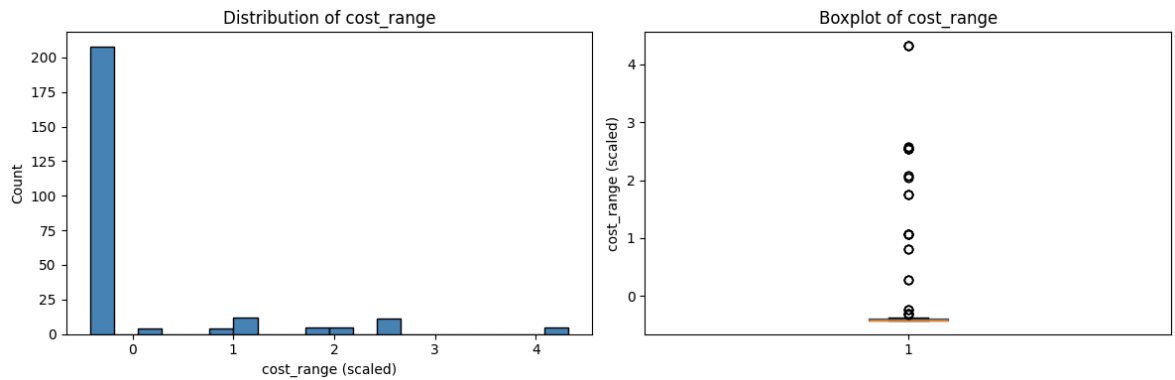
```
In [332... # Do not modify this code
print_tile(size='h3', key='feature_6_insights', value=feature_6_insights)

feature_6_insights
```

latest_standard_cost is the most recent production cost. Together with latest_list_price it is one of the two inputs to estimated_margin_pct. Pre-scaling distribution: range \$0.86–\$2,171.29, median \$200, mean \$442, std \$557. Heavily right-skewed — most products have relatively low production costs while a minority of complex, high-value products carry very high costs. latest_standard_cost is correlated with latest_list_price (r=0.90 in SD-a E.4) but was retained because cost structure adds information beyond price level: two products with the same list price can have very different cost bases and therefore very different margin profiles. The correlation did not meet the 0.85 drop threshold on the pair-wise check, and cost is directionally distinct from price in the margin calculation.

C.12 Explore Feature of Interest `cost_range`

```
In [333... fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(X_train['cost_range'], bins=20, edgecolor='black', color='st
axes[0].set_xlabel('cost_range (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of cost_range')
axes[1].boxplot(X_train['cost_range'].dropna(), vert=True, patch_artist=T
                boxprops=dict(facecolor='steelblue', alpha=0.6))
axes[1].set_ylabel('cost_range (scaled)')
axes[1].set_title('Boxplot of cost_range')
plt.tight_layout()
plt.show()
print(X_train['cost_range'].describe().round(3))
print(f'Products with zero cost history change: {(X_train["cost_range"] =
```



```
count    254.000
mean      0.000
std       1.002
min      -0.422
25%      -0.422
50%      -0.422
75%      -0.393
max       4.323
Name: cost_range, dtype: float64
Products with zero cost history change: 187
```

```
In [334...] feature_7_insights = """
cost_range is the difference between the highest and lowest historical st
recorded for each product. It measures cost volatility over the product l

Pre-scaling: 187 products (74%) have a cost range of \$0 – their producti
has never changed. Only 67 products show any cost variation, with a maxim
of \$234.26. The spike at zero dominates the scaled distribution.

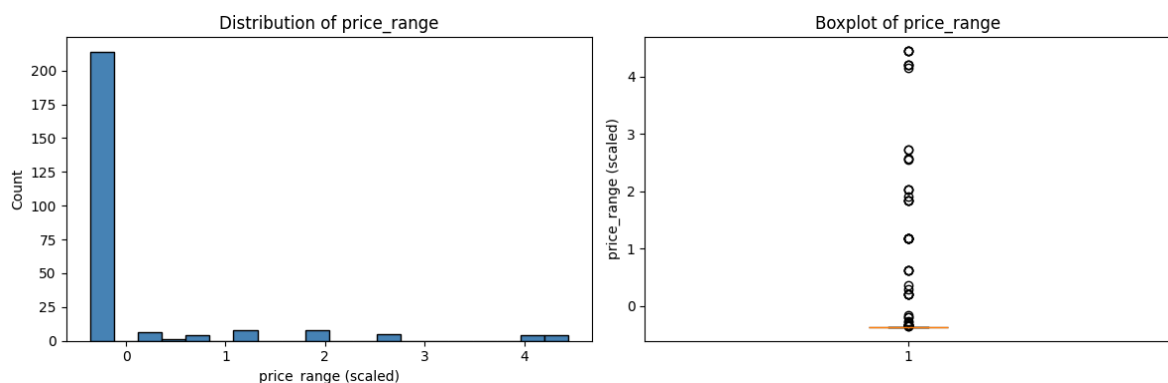
A non-zero cost_range indicates that a product's production cost has been
at least once – due to input cost changes, process changes, or revaluatio
with large cost ranges have experienced significant cost instability; if
did not adjust proportionally, margin will have compressed. Like max_disc
this feature has a bimodal structure after scaling: a large spike at the
(zero-change products) and a spread of values for the volatile minority.
"""
```

```
In [335...] # Do not modify this code
print_tile(size='h3', key='feature_7_insights', value=feature_7_insights)
```

cost_range is the difference between the highest and lowest historical standard cost recorded for each product. It measures cost volatility over the product lifetime. Pre-scaling: 187 products (74%) have a cost range of \$0 — their production cost has never changed. Only 67 products show any cost variation, with a maximum range of \$234.26. The spike at zero dominates the scaled distribution. A non-zero **cost_range** indicates that a product's production cost has been revised at least once — due to input cost changes, process changes, or revaluation. Products with large cost ranges have experienced significant cost instability; if list prices did not adjust proportionally, margin will have compressed. Like **max_discount_pct**, this feature has a bimodal structure after scaling: a large spike at the minimum (zero-change products) and a spread of values for the volatile minority.

C.13 Explore Feature of Interest **price_range**

```
In [336... fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(X_train['price_range'], bins=20, edgecolor='black', color='s
axes[0].set_xlabel('price_range (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of price_range')
axes[1].boxplot(X_train['price_range'].dropna(), vert=True, patch_artist=
                boxprops=dict(facecolor='steelblue', alpha=0.6))
axes[1].set_ylabel('price_range (scaled)')
axes[1].set_title('Boxplot of price_range')
plt.tight_layout()
plt.show()
print(X_train['price_range'].describe().round(3))
print(f'Products with no price change: {(X_train["price_range"] == X_trai
```



```

count      254.000
mean        0.000
std         1.002
min         -0.363
25%         -0.363
50%         -0.363
75%         -0.363
max          4.444
Name: price_range, dtype: float64
Products with no price change: 200

```

```

In [337... feature_8_insights = """
price_range is the difference between the highest and lowest historical list price
recorded for each product. It measures price stability over the product lifetime.

Pre-scaling: 200 products in X_train sit at the minimum scaled value (corresponding
to a $0 price range), meaning their list price has never changed. Products with
non-zero price_range have been repriced at least once, with a maximum range of
$261.79. Price stability signals steady-state demand with predictable margin dynamics.
A high price_range can indicate promotional pricing (price cut and restored) or a
permanent reprice under market pressure. Products with both high price_range and
high max_discount_pct have been subjected to pricing pressure from multiple
directions simultaneously — the strongest compound margin-risk signal available in
this dataset.
"""

```

```

In [338... # Do not modify this code
print_tile(size='h3', key='feature_8_insights', value=feature_8_insights)

```

feature_8_insights

price_range is the difference between the highest and lowest historical list price recorded for each product. It measures price stability over the product lifetime. Pre-scaling: 200 products in X_train sit at the minimum scaled value (corresponding to a \$0 price range), meaning their list price has never changed. Products with non-zero price_range have been repriced at least once, with a maximum range of \$261.79. Price stability signals steady-state demand with predictable margin dynamics. A high price_range can indicate promotional pricing (price cut and restored) or a permanent reprice under market pressure. Products with both high price_range and high max_discount_pct have been subjected to pricing pressure from multiple directions simultaneously — the strongest compound margin-risk signal available in this dataset.

C.14 Explore Feature of Interest total_quantity_sold

```

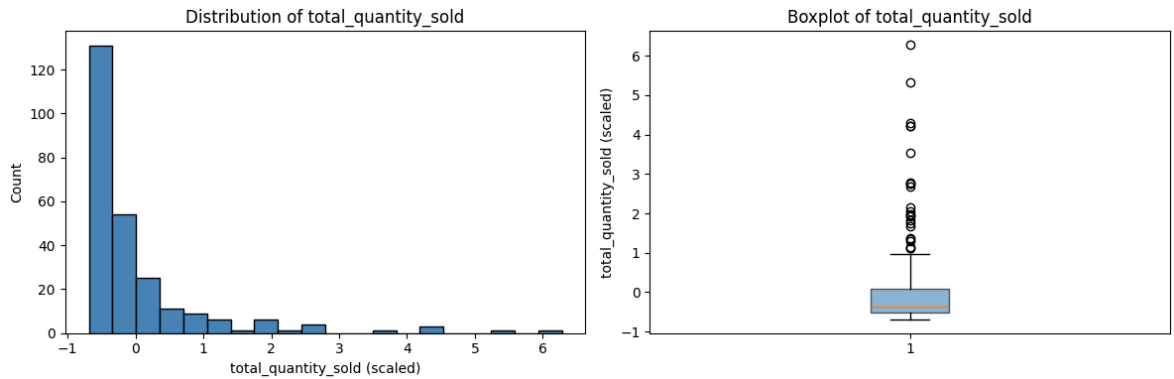
In [339... fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(X_train['total_quantity_sold'], bins=20, edgecolor='black',
axes[0].set_xlabel('total_quantity_sold (scaled)')
axes[0].set_ylabel('Count')
axes[0].set_title('Distribution of total_quantity_sold')
axes[1].boxplot(X_train['total_quantity_sold'].dropna(), vert=True, patch
boxprops=dict(facecolor='steelblue', alpha=0.6))

```

```

axes[1].set_ylabel('total_quantity_sold (scaled)')
axes[1].set_title('Boxplot of total_quantity_sold')
plt.tight_layout()
plt.show()
print(X_train['total_quantity_sold'].describe().round(3))

```



```

count      254.000
mean        0.000
std         1.002
min        -0.689
25%        -0.526
50%        -0.366
75%         0.085
max         6.281
Name: total_quantity_sold, dtype: float64

```

In [340...

```

feature_9_insights = """
total_quantity_sold is the cumulative units sold across all transactions
product. It measures demand volume over the full sales history.

Pre-scaling distribution: range 1-2,837 units, median 132, mean 281, std
Heavily right-skewed – a small number of products account for very high c
volumes while the majority sit well below the mean.

Volume is critical for margin-risk prioritisation: high-volume products w
margins carry the greatest aggregate financial exposure – small per-unit
changes multiply across a large quantity base. Low-volume products with t
are less impactful in absolute terms but may signal early demand problems
combining high volume with high estimated_margin_pct identifies commercia
products that should be protected from discount pressure.
"""

```

In [341...

```

# Do not modify this code
print_tile(size='h3', key='feature_9_insights', value=feature_9_insights)

```

total_quantity_sold is the cumulative units sold across all transactions for each product. It measures demand volume over the full sales history. Pre-scaling distribution: range 1–2,837 units, median 132, mean 281, std 408. Heavily right-skewed — a small number of products account for very high cumulative volumes while the majority sit well below the mean. Volume is critical for margin-risk prioritisation: high-volume products with thin margins carry the greatest aggregate financial exposure — small per-unit margin changes multiply across a large quantity base. Low-volume products with thin margins are less impactful in absolute terms but may signal early demand problems. A cluster combining high volume with high estimated_margin_pct identifies commercial anchors — products that should be protected from discount pressure.

C.15 Explore Feature of Interest `num_discount_offers`

```
In [342... # 5 distinct raw values: 0-4 – plot as labelled bar chart
raw_counts_ndo = {0: 112, 1: 75, 2: 45, 3: 17, 4: 5}
fig, ax = plt.subplots(figsize=(7, 4))
bars = ax.bar([str(k) for k in sorted(raw_counts_ndo)],
              [raw_counts_ndo[k] for k in sorted(raw_counts_ndo)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [raw_counts_ndo[k] for k in sorted(raw_counts_ndo)]:
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_xlabel('Number of discount offers (original value)')
ax.set_ylabel('Count')
ax.set_title('Product count by num_discount_offers')
plt.tight_layout()
plt.show()
print(X_train['num_discount_offers'].describe().round(3))
```




```

count      254.000
mean       -0.000
std         1.002
min        -0.903
25%        -0.903
50%         0.069
75%         1.041
max         2.986
Name: num_discount_offers, dtype: float64

```

```

In [343... feature_10_insights = """
num_discount_offers counts the distinct real promotional offers ever asso
with each product (No Discount placeholders excluded).

Distribution: 112 products (44%) have never been included in a real disco
75 have been included in exactly 1; 45 in 2; 17 in 3; 5 in 4. The feature
takes only 5 unique values, making it more categorical than continuous in

Together with max_discount_pct and has_volume_discount, this feature form
promotion exposure profile. A high num_discount_offers combined with high
max_discount_pct indicates a product that has been repeatedly and deeply
the clearest signal of sustained demand support risk in this dataset.
"""

```

```

In [344... # Do not modify this code
print_tile(size='h3', key='feature_10_insights', value=feature_10_insight

feature_10_insights

```

num_discount_offers counts the distinct real promotional offers ever associated with each product (No Discount placeholders excluded). Distribution: 112 products (44%) have never been included in a real discount offer; 75 have been included in exactly 1; 45 in 2; 17 in 3; 5 in 4. The feature effectively takes only 5 unique values, making it more categorical than continuous in practice. Together with max_discount_pct and has_volume_discount, this feature forms the promotion exposure profile. A high num_discount_offers combined with high max_discount_pct indicates a product that has been repeatedly and deeply discounted — the clearest signal of sustained demand support risk in this dataset.

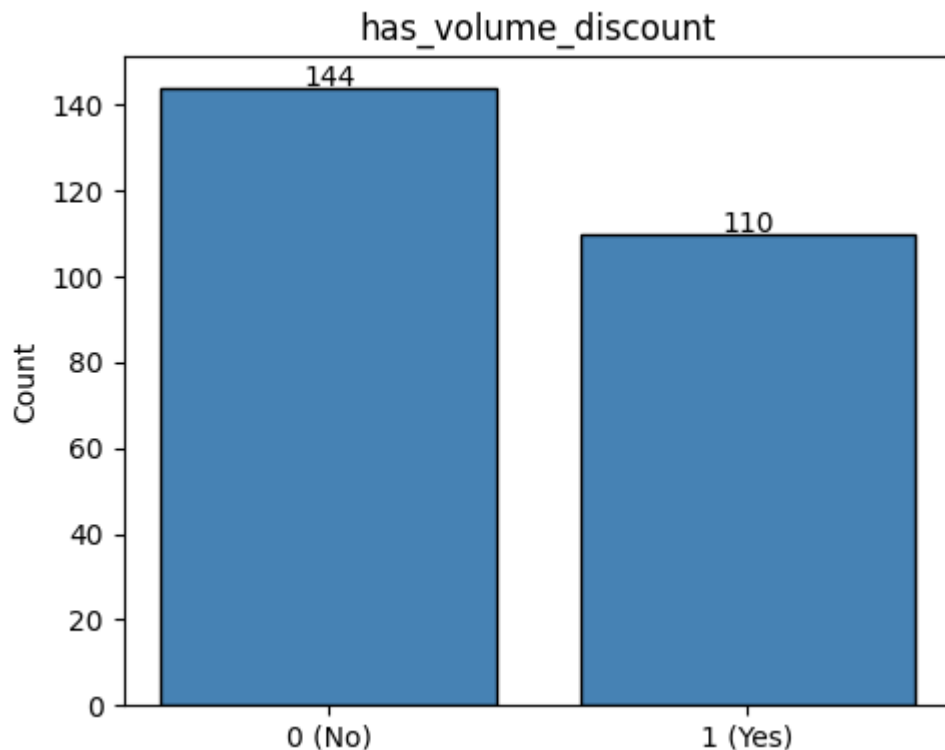
C.16 Explore Feature of Interest `has_volume_discount`

```

In [345... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['has_volume_discount'].value_counts().sort_index()
bars = ax.bar(['0 (No)', '1 (Yes)'], [counts.get(0,0), counts.get(1,0)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('has_volume_discount')
plt.tight_layout()

```

```
plt.show()
print(X_train['has_volume_discount'].value_counts())
```



```
has_volume_discount
0.0    144
1.0    110
Name: count, dtype: int64
```

```
In [346... feature_11_insights = """
has_volume_discount is a binary flag: 1 = product has been in a volume di
offer (110 products, 43%), 0 = it has not (144 products, 57%).

Volume discounts are a deliberate pricing strategy – they trade reduced p
margin for higher throughput, targeting bulk buyers. This is structurally
from clearance discounting.

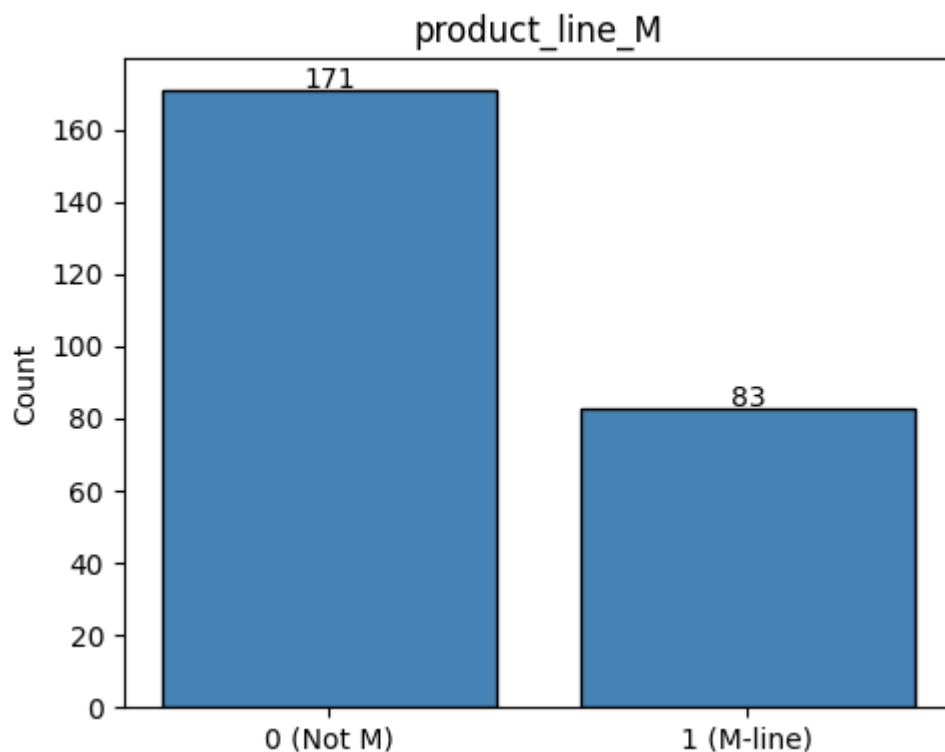
Products with has_volume_discount=1 and strong total_quantity_sold repres
volume-strategy cluster: intentional margin trade-off for scale. Products
has_volume_discount=1 but low total_quantity_sold may indicate the discou
failed to generate the intended uplift – a potential margin-risk signal i
discounts were also deep (high max_discount_pct).
"""
```

```
In [347... # Do not modify this code
print_tile(size='h3', key='feature_11_insights', value=feature_11_insight
```

`has_volume_discount` is a binary flag: 1 = product has been in a volume discount offer (110 products, 43%), 0 = it has not (144 products, 57%). Volume discounts are a deliberate pricing strategy — they trade reduced per-unit margin for higher throughput, targeting bulk buyers. This is structurally different from clearance discounting. Products with `has_volume_discount=1` and strong `total_quantity_sold` represent a volume-strategy cluster: intentional margin trade-off for scale. Products with `has_volume_discount=1` but low `total_quantity_sold` may indicate the discount failed to generate the intended uplift — a potential margin-risk signal if the discounts were also deep (high `max_discount_pct`).

C.17 Explore Feature of Interest `product_line_M`

```
In [348... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['product_line_M'].value_counts().sort_index()
bars = ax.bar(['0 (Not M)', '1 (M-line)'], [counts.get(0,0), counts.get(1,0)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('product_line_M')
plt.tight_layout()
plt.show()
print(X_train['product_line_M'].value_counts())
```



```
product_line_M
0    171
1     83
Name: count, dtype: int64
```

```
In [349... feature_12_insights = """
product_line_M is the one-hot indicator for the Mountain line. 83 product
are Mountain; 171 are not.

Mountain products cover off-road cycling equipment with higher manufactur
complexity, longer production times, and higher standard costs. They are
predominantly in-house manufactured and skew toward higher class tiers.
As a one-hot feature, product_line_M contributes a binary distance signal
two products are either both M-line (distance 0) or one is M-line and the
other is not (distance 1) on this dimension.
"""
```

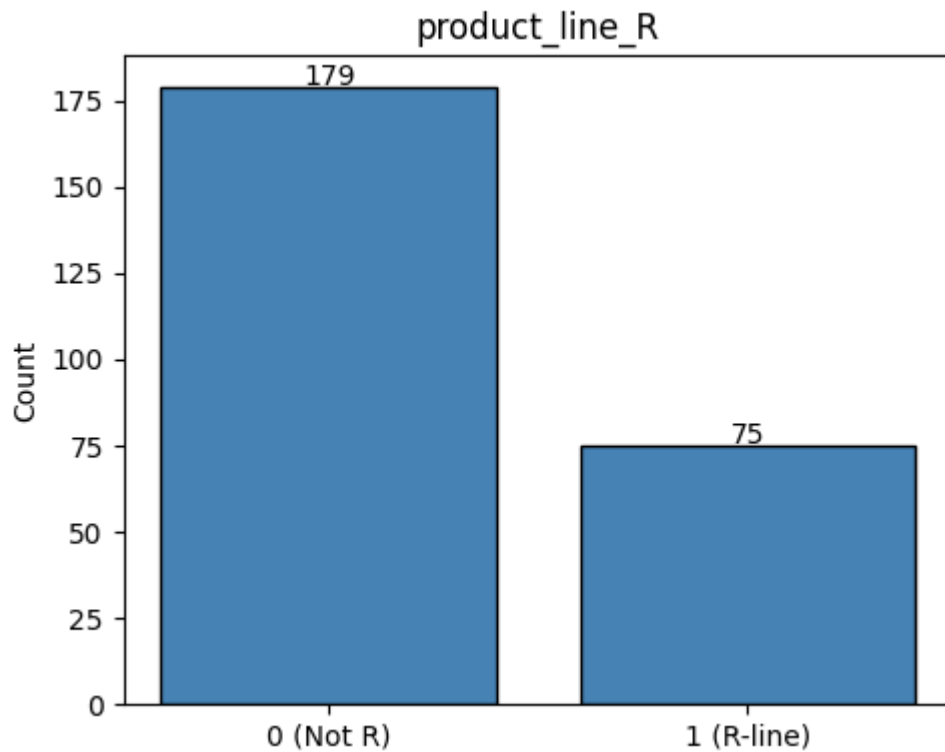
```
In [350... # Do not modify this code
print_tile(size='h3', key='feature_12_insights', value=feature_12_insight
```

feature_12_insights

product_line_M is the one-hot indicator for the Mountain line. 83 products (33%) are Mountain; 171 are not. Mountain products cover off-road cycling equipment with higher manufacturing complexity, longer production times, and higher standard costs. They are predominantly in-house manufactured and skew toward higher class tiers. As a one-hot feature, product_line_M contributes a binary distance signal: two products are either both M-line (distance 0) or one is M-line and the other is not (distance 1) on this dimension.

C.18 Explore Feature of Interest product_line_R

```
In [351... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['product_line_R'].value_counts().sort_index()
bars = ax.bar(['0 (Not R)', '1 (R-line)'], [counts.get(0,0), counts.get(1,0)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('product_line_R')
plt.tight_layout()
plt.show()
print(X_train['product_line_R'].value_counts())
```



```
product_line_R
0    179
1     75
Name: count, dtype: int64
```

```
In [352... feature_13_insights = """
product_line_R is the one-hot indicator for the Road line. 75 products (30%)
are Road; 179 are not.

Road products span the widest list price range in the portfolio – from an
entry-level to premium performance models. R-line products may therefore
appear in multiple clusters based on price tier and margin profile, rather than
forming a single cohesive cluster on product line alone. R-line and M-line
together account for 63% of the 254 training products.
"""
```

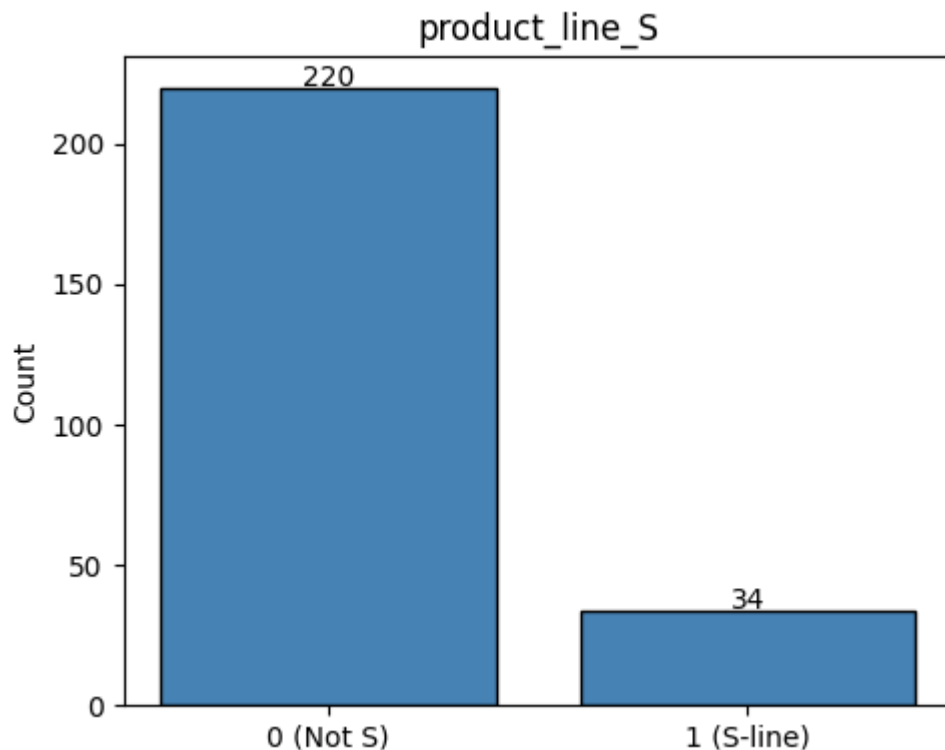
```
In [353... # Do not modify this code
print_tile(size='h3', key='feature_13_insights', value=feature_13_insights)
```

feature_13_insights

product_line_R is the one-hot indicator for the Road line. 75 products (30%) are Road; 179 are not. Road products span the widest list price range in the portfolio — from affordable entry-level to premium performance models. R-line products may therefore appear in multiple clusters based on price tier and margin profile, rather than forming a single cohesive cluster on product line alone. R-line and M-line together account for 63% of the 254 training products.

C.19 Explore Feature of Interest `product_line_S`

```
In [354... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['product_line_S'].value_counts().sort_index()
bars = ax.bar(['0 (Not S)', '1 (S-line)'], [counts.get(0,0), counts.get(1,0)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('product_line_S')
plt.tight_layout()
plt.show()
print(X_train['product_line_S'].value_counts())
```



```
product_line_S
0    220
1     34
Name: count, dtype: int64
```

```
In [355... feature_14_insights = """
product_line_S is the one-hot indicator for the Specialty line. 34 products
are Specialty; 220 are not.

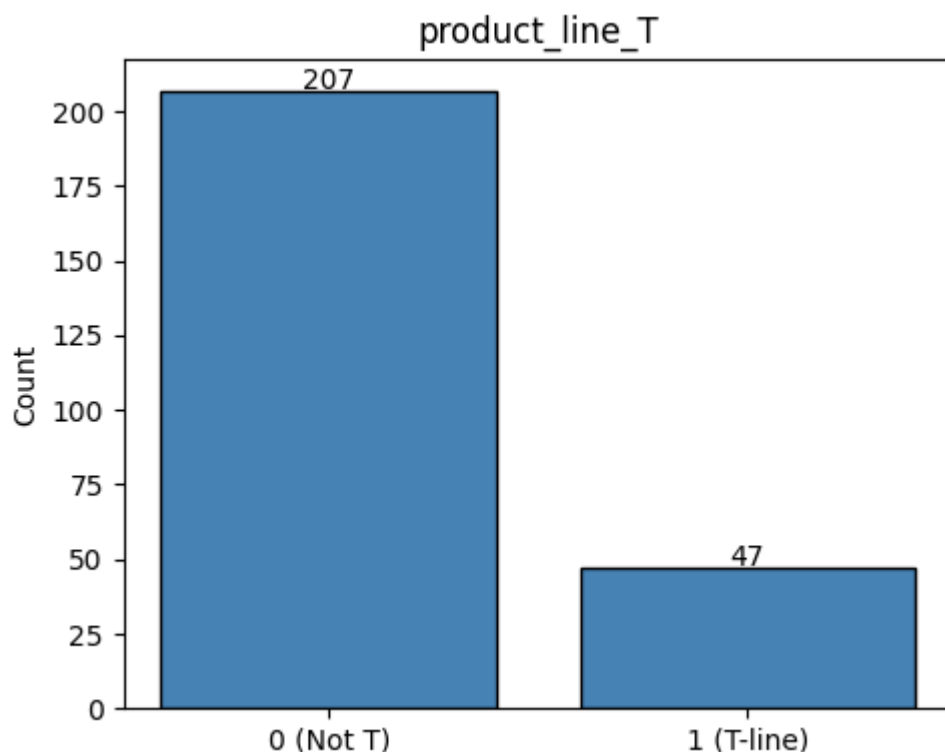
Specialty products are niche items (accessories, clothing) with simple market
profiles – predominantly purchased and carrying lower standard costs. The
prevalence means S-line membership is less likely to be a primary cluster
but may contribute to defining a low-cost, low-price accessory cluster.
"""
```

```
In [356... # Do not modify this code
print_tile(size='h3', key='feature_14_insights', value=feature_14_insights)
```

product_line_S is the one-hot indicator for the Specialty line. 34 products (13%) are Specialty; 220 are not. Specialty products are niche items (accessories, clothing) with simple manufacturing profiles — predominantly purchased and carrying lower standard costs. The lower prevalence means S-line membership is less likely to be a primary cluster differentiator, but may contribute to defining a low-cost, low-price accessory cluster.

C.20 Explore Feature of Interest product_line_T

```
In [357... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['product_line_T'].value_counts().sort_index()
bars = ax.bar(['0 (Not T)', '1 (T-line)'], [counts.get(0,0), counts.get(1,0)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('product_line_T')
plt.tight_layout()
plt.show()
print(X_train['product_line_T'].value_counts())
```



```
product_line_T
0    207
1     47
Name: count, dtype: int64
```

```
In [358... feature_15_insights = """
product_line_T is the one-hot indicator for the Touring line. 47 products
are Touring; 207 are not.
```

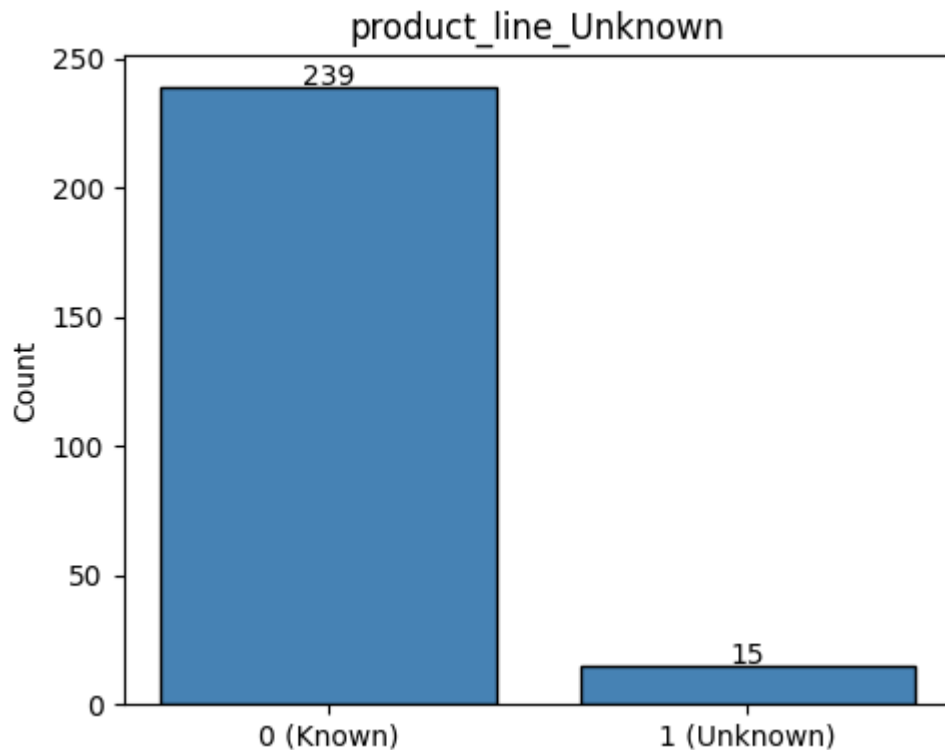
Touring products are designed for endurance riding, sitting between Mount Specialty in price and complexity. T-line products tend to sit in the mid range with moderate manufacturing lead times and more stable margin profile as the touring segment attracts less aggressive promotional pricing than Mountain.

```
In [359... # Do not modify this code
print_tile(size='h3', key='feature_15_insights', value=feature_15_insight
feature_15_insights
```

product_line_T is the one-hot indicator for the Touring line. 47 products (19%) are Touring; 207 are not. Touring products are designed for endurance riding, sitting between Mountain and Specialty in price and complexity. T-line products tend to sit in the mid-price range with moderate manufacturing lead times and more stable margin profiles, as the touring segment attracts less aggressive promotional pricing than Mountain.

C.21 Explore Feature of Interest product_line_Unknown

```
In [360... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['product_line_Unknown'].value_counts().sort_index()
bars = ax.bar(['0 (Known)', '1 (Unknown)'], [counts.get(0,0), counts.get(
    edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('product_line_Unknown')
plt.tight_layout()
plt.show()
print(X_train['product_line_Unknown'].value_counts())
```

```
product_line_Unknown
0      239
1       15
Name: count, dtype: int64
```

```
In [361...] feature_16_insights = """
product_line_Unknown is the one-hot indicator for products whose product_
was missing and imputed as 'Unknown' in SD-a E.1. 15 products (6%) fall h

Rather than imputing a modal product line, the Unknown category preserves
data gap as a distinct value, allowing the algorithm to determine whether
15 products share characteristics with a named line or form a distinct gr
With 6% prevalence, this group sits just above the 5.5% variance threshol
that removed style_M (~2%), and was therefore retained.
"""
```

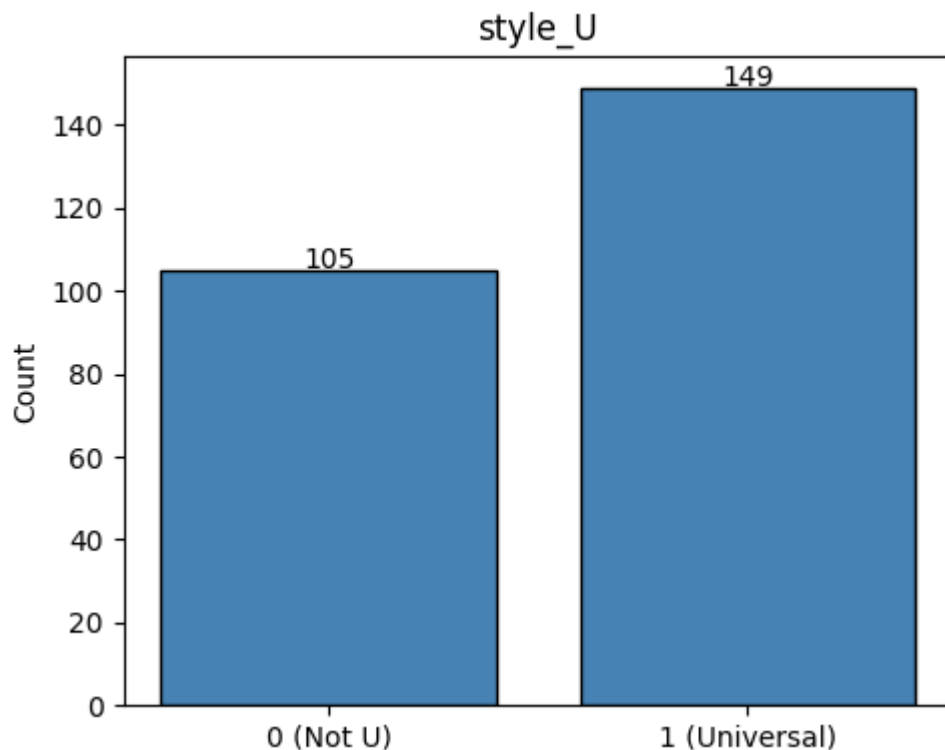
```
In [362...] # Do not modify this code
print_tile(size='h3', key='feature_16_insights', value=feature_16_insight
```

feature_16_insights

product_line_Unknown is the one-hot indicator for products whose product_line was missing and imputed as 'Unknown' in SD-a E.1. 15 products (6%) fall here. Rather than imputing a modal product line, the Unknown category preserves the data gap as a distinct value, allowing the algorithm to determine whether these 15 products share characteristics with a named line or form a distinct group. With 6% prevalence, this group sits just above the 5.5% variance threshold that removed style_M (~2%), and was therefore retained.

C.22 Explore Feature of Interest `style_U`

```
In [363... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['style_U'].value_counts().sort_index()
bars = ax.bar(['0 (Not U)', '1 (Universal)'], [counts.get(0,0), counts.get(1,0)],
              edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('style_U')
plt.tight_layout()
plt.show()
print(X_train['style_U'].value_counts())
```



```
style_U
1    149
0    105
Name: count, dtype: int64
```

```
In [364... feature_17_insights = """
style_U is the one-hot indicator for Universal style. 149 products (59%)
Universal style; 105 do not.

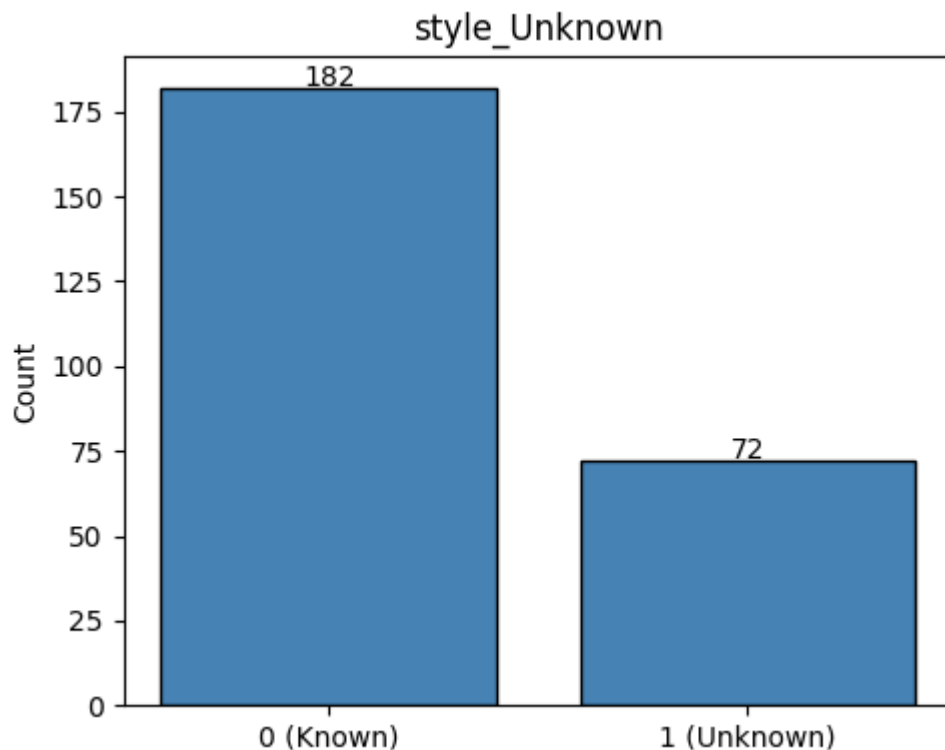
Universal-style products have no gender-specific sizing, representing acc
components, and equipment that apply to all riders. This is the dominant
category, reflecting that most of the portfolio consists of components ra
than apparel. Given 59% prevalence, style_U will be the modal style in ma
clusters; style-driven cluster separation is more likely to be driven by
commercial features.
"""
```

```
In [365... # Do not modify this code
print_tile(size='h3', key='feature_17_insights', value=feature_17_insight
```

style_U is the one-hot indicator for Universal style. 149 products (59%) have Universal style; 105 do not. Universal-style products have no gender-specific sizing, representing accessories, components, and equipment that apply to all riders. This is the dominant style category, reflecting that most of the portfolio consists of components rather than apparel. Given 59% prevalence, style_U will be the modal style in many clusters; style-driven cluster separation is more likely to be driven by other commercial features.

C.23 Explore Feature of Interest style_Unknown

```
In [366... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['style_Unknown'].value_counts().sort_index()
bars = ax.bar(['0 (Known)', '1 (Unknown)'], [counts.get(0,0), counts.get(
    edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('style_Unknown')
plt.tight_layout()
plt.show()
print(X_train['style_Unknown'].value_counts())
```



```
style_Unknown
0    182
1     72
Name: count, dtype: int64
```

```
In [367... feature_18_insights = """
style_Unknown is the one-hot indicator for products whose style was missi
```

```
imputed as 'Unknown' in SD-a E.1. 72 products (28%) fall here.
```

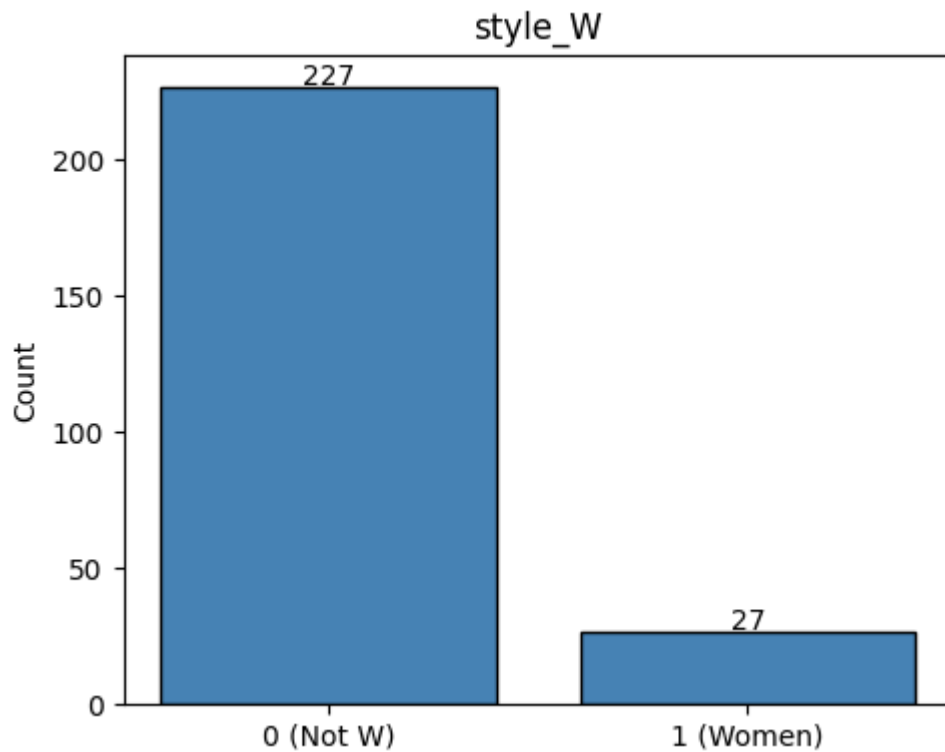
```
Unknown style is the second-largest style group after Universal. Products
Unknown style are predominantly components and accessories where style is
relevant attribute, which explains the high rate of missing values. With
prevalence, this group is large enough to potentially anchor its own clus
or appear prominently across multiple clusters.
"""
```

```
In [368... # Do not modify this code
print_tile(size='h3', key='feature_18_insights', value=feature_18_insight
feature_18_insights
```

style_Unknown is the one-hot indicator for products whose style was missing and imputed as 'Unknown' in SD-a E.1. 72 products (28%) fall here. Unknown style is the second-largest style group after Universal. Products with Unknown style are predominantly components and accessories where style is not a relevant attribute, which explains the high rate of missing values. With 28% prevalence, this group is large enough to potentially anchor its own cluster or appear prominently across multiple clusters.

C.24 Explore Feature of Interest style_W

```
In [369... fig, ax = plt.subplots(figsize=(5, 4))
counts = X_train['style_W'].value_counts().sort_index()
bars = ax.bar(['0 (Not W)', '1 (Women)'], [counts.get(0,0), counts.get(1,
    edgecolor='black', color='steelblue')
for bar, v in zip(bars, [counts.get(0,0), counts.get(1,0)]):
    ax.text(bar.get_x() + bar.get_width()/2, v + 0.5, str(v), ha='center')
ax.set_ylabel('Count')
ax.set_title('style_W')
plt.tight_layout()
plt.show()
print(X_train['style_W'].value_counts())
```



```
style_W
0      227
1       27
Name: count, dtype: int64
```

In [370... `feature_19_insights = """`
`style_W is the one-hot indicator for Women-specific style. 27 products (11%) have Women style; 227 do not.`

`Women-specific products are mostly apparel and accessories (jerseys, shorts, gloves) with relatively low list prices and simple manufacturing profiles. With 11% prevalence, style_W is above the variance threshold that removed style_M (~2%) and was retained. It represents a minority signal – any cluster where W-style products concentrate will be a relatively small group but commercially distinct (low price, low cost, apparel-focused).`
`"""`

In [371... `# Do not modify this code`
`print_tile(size='h3', key='feature_19_insights', value=feature_19_insights)`

feature_19_insights

style_W is the one-hot indicator for Women-specific style. 27 products (11%) have Women style; 227 do not. Women-specific products are mostly apparel and accessories (jerseys, shorts, gloves) with relatively low list prices and simple manufacturing profiles. With 11% prevalence, style_W is above the variance threshold that removed style_M (~2%) and was retained. It represents a minority signal — any cluster where W-style products concentrate will be a relatively small group but commercially distinct (low price, low cost, apparel-focused).

D. Feature Selection

```
In [372... features_list = [  
    # Structural / product attributes  
    'class', 'days_to_manufacture', 'is_manufactured',  
    # Cost and pricing  
    'latest_standard_cost', 'cost_range', 'latest_list_price', 'price_range',  
    # Sales behaviour  
    'total_quantity_sold',  
    # Promotion exposure  
    'num_discount_offers', 'max_discount_pct', 'has_volume_discount',  
    # Margin  
    'estimated_margin_pct',  
    # Product line (one-hot)  
    'product_line_M', 'product_line_R', 'product_line_S', 'product_line_T',  
    # Style (one-hot, style_M excluded at variance filter)  
    'style_U', 'style_Unknown', 'style_W',  
]  
print(f'Features selected: {len(features_list)}')  
print(features_list)
```

Features selected: 20

```
['class', 'days_to_manufacture', 'is_manufactured', 'latest_standard_cost',  
'cost_range', 'latest_list_price', 'price_range', 'total_quantity_sold',  
'num_discount_offers', 'max_discount_pct', 'has_volume_discount', 'estimated_margin_pct',  
'product_line_M', 'product_line_R', 'product_line_S', 'product_line_T',  
'product_line_Unknown', 'style_U', 'style_Unknown', 'style_W']
```

In [372...

```
In [373... feature_selection_explanations = """  
20 features are carried forward into the clustering stage. All selection  
were made and documented in the data preparation notebook (SD-a); this set  
the final feature set that X_train.csv contains.  
  
Features are organised across four commercial dimensions:  
- Structural / product attributes: class, days_to_manufacture, is_manuf  
- Cost and pricing: latest_standard_cost, cost_range, latest_list_price  
- Sales behaviour: total_quantity_sold (num_orders was dropped, r=0.91  
- Promotion exposure: num_discount_offers, max_discount_pct, has_volume  
- Margin: estimated_margin_pct (estimated_margin dropped, r=0.99 with l  
- Category membership: product_line (M/R/S/T/Unknown), style (U/W/Unkno  
  
Two exclusions applied during SD-a:  
- style_M was dropped: variance 0.023, corresponding to ~2% prevalence  
A category present in fewer than ~6 products cannot meaningfully anch  
- estimated_margin and num_orders were dropped due to near-perfect corr  
latest_list_price (r=0.99) and total_quantity_sold (r=0.91) respectiv  
correlated features would cause those dimensions to contribute double  
Euclidean distances in all three clustering algorithms.  
"""
```

```
In [374... # Do not modify this code  
print_tile(size="h3", key='feature_selection_explanations', value=feature
```

20 features are carried forward into the clustering stage. All selection decisions were made and documented in the data preparation notebook (SD-a); this section confirms the final feature set that X_train.csv contains. Features are organised across four commercial dimensions: - Structural / product attributes: class, days_to_manufacture, is_manufactured - Cost and pricing: latest_standard_cost, cost_range, latest_list_price, price_range - Sales behaviour: total_quantity_sold (num_orders was dropped, $r=0.91$ with total_quantity_sold) - Promotion exposure: num_discount_offers, max_discount_pct, has_volume_discount - Margin: estimated_margin_pct (estimated_margin dropped, $r=0.99$ with latest_list_price) - Category membership: product_line (M/R/S/T/Unknown), style (U/W/Unknown) as one-hot columns Two exclusions applied during SD-a: - style_M was dropped: variance 0.023, corresponding to $\sim 2\%$ prevalence across 254 products. A category present in fewer than ~ 6 products cannot meaningfully anchor a cluster boundary. - estimated_margin and num_orders were dropped due to near-perfect correlation with latest_list_price ($r=0.99$) and total_quantity_sold ($r=0.91$) respectively. Retaining correlated features would cause those dimensions to contribute double weight to Euclidean distances in all three clustering algorithms.

E. Data Preparation

E.1 Imputation (completed in SD-a)

```
In [375... # All imputation was completed in SD-a E.1.
# X_train.csv already contains the imputed values; no further action is t
```

```
In [375...
```

```
In [376... data_cleaning_1_explanations = """
All missing-value imputation was performed in the data preparation notebo
and the results are persisted in X_train.csv. Repeating it here would be
and could produce different fill values if the source tables had changed.

The imputation strategy applied in SD-a:
- product_line and style: missing values filled with 'Unknown', treated
  distinct nominal category. A separate category is preferable to imput
  mode because missing product_line may indicate a genuinely uncategori
  rather than a data entry error.
- class: missing values filled with the mode ('M'). Class is ordinal (L
  'Unknown' cannot be placed on the ordinal scale, so mode imputation p
  the meaningful distance structure required for ordinal encoding.
- Numeric features with nulls (latest_list_price, price_range, estimate
```

```
estimated_margin_pct): imputed with the column median to minimise the  
of the right-skewed price distribution on the fill value.  
"""
```

```
In [377... # Do not modify this code  
print_tile(size="h3", key='data_cleaning_1_explanations', value=data_clea  
  
data_cleaning_1_explanations
```

All missing-value imputation was performed in the data preparation notebook (SD-a) and the results are persisted in X_train.csv. Repeating it here would be redundant and could produce different fill values if the source tables had changed. The imputation strategy applied in SD-a: - product_line and style: missing values filled with 'Unknown', treated as a distinct nominal category. A separate category is preferable to imputing a mode because missing product_line may indicate a genuinely uncategorised product rather than a data entry error. - class: missing values filled with the mode ('M'). Class is ordinal (L/M/H) and 'Unknown' cannot be placed on the ordinal scale, so mode imputation preserves the meaningful distance structure required for ordinal encoding. - Numeric features with nulls (latest_list_price, price_range, estimated_margin, estimated_margin_pct): imputed with the column median to minimise the influence of the right-skewed price distribution on the fill value.

E.2 Encoding (completed in SD-a)

```
In [378... # All encoding was completed in SD-a E.2.  
# X_train.csv already contains encoded columns; no further action is take
```

```
In [378...
```

```
In [379... data_cleaning_2_explanations = """  
All categorical encoding was performed in SD-a and is already reflected i  
  
Encoding decisions from SD-a:  
- class (L/M/H): ordinal encoding {L:0, M:1, H:2}. Class has a natural  
by production complexity, so the numeric positions preserve meaningfu  
An L-class product is one step from M and two steps from H, which cor  
represents their relative difference under Euclidean distance.  
- product_line and style: one-hot encoding with drop_first=False. Both  
with no natural ordering between categories. Dropping a reference cat  
create asymmetric Euclidean distances, where the dropped category app  
from all others – a geometric distortion that penalises clustering al  
that rely on centroids or pairwise distances.  
"""
```

```
In [380... # Do not modify this code  
print_tile(size="h3", key='data_cleaning_2_explanations', value=data_clea
```


All categorical encoding was performed in SD-a and is already reflected in X_train.csv. Encoding decisions from SD-a: - class (L/M/H): ordinal encoding {L:0, M:1, H:2}. Class has a natural ordering by production complexity, so the numeric positions preserve meaningful distances. An L-class product is one step from M and two steps from H, which correctly represents their relative difference under Euclidean distance. - product_line and style: one-hot encoding with drop_first=False. Both are nominal with no natural ordering between categories. Dropping a reference category would create asymmetric Euclidean distances, where the dropped category appears equidistant from all others — a geometric distortion that penalises clustering algorithms that rely on centroids or pairwise distances.

E.3 Scaling and Variance / Correlation Filtering (completed in SD-a)

```
In [381... # All scaling and feature filtering was completed in SD-a E.3 and E.4.
# X_train.csv already contains scaled, filtered features; no further acti
```

```
In [381...
```

```
In [382... data_cleaning_3_explanations = """
Scaling and feature filtering were applied in SD-a (E.3 and E.4) before X

Scaling (SD-a E.3): StandardScaler was applied to all continuous and ordi
Binary features (is_manufactured, has_volume_discount, all one-hot column
from scaling – their 0/1 values already occupy a bounded, comparable rang
would compress their contribution to distances rather than standardise it

Variance filtering (SD-a E.4): style_M was removed (variance 0.023, <5.5%
A binary feature at that prevalence level represents roughly 5 products a
cannot meaningfully anchor a cluster boundary.

Correlation filtering (SD-a E.4): estimated_margin (r=0.994 with latest_l
num_orders (r=0.913 with total_quantity_sold) were dropped. All three clu
used here (KMeans, Mean Shift, Agglomerative) compute Euclidean distances
highly correlated features would cause those dimensions to contribute dis
weight to cluster shapes.
"""
```

```
In [383... # Do not modify this code
print_tile(size="h3", key='data_cleaning_3_explanations', value=data_clea
```

Scaling and feature filtering were applied in SD-a (E.3 and E.4) before X_train.csv was saved. Scaling (SD-a E.3): StandardScaler was applied to all continuous and ordinal features. Binary features (is_manufactured, has_volume_discount, all one-hot columns) were excluded from scaling — their 0/1 values already occupy a bounded, comparable range, and scaling would compress their contribution to distances rather than standardise it. Variance filtering (SD-a E.4): style_M was removed (variance 0.023, 5.5% prevalence). A binary feature at that prevalence level represents roughly 5 products across 254 and cannot meaningfully anchor a cluster boundary. Correlation filtering (SD-a E.4): estimated_margin ($r=0.994$ with latest_list_price) and num_orders ($r=0.913$ with total_quantity_sold) were dropped. All three clustering algorithms used here (KMeans, Mean Shift, Agglomerative) compute Euclidean distances, so retaining highly correlated features would cause those dimensions to contribute disproportionate weight to cluster shapes.

F. Feature Engineering

F.1 Sales Behaviour Features (completed in SD-a)

In [384... *# Sales behaviour features were engineered in SD-a D.1 and are present in*

In [384...

In [385... feature_engineering_1_explanations = """
 Two sales behaviour features were derived in SD-a D.1 by aggregating the table at the product level:

- total_quantity_sold: sum of order_quantity across all transactions for High volume with low price is a distinct commercial profile from high high price, and this feature anchors that distinction.
- num_orders: count of distinct sales orders containing the product. This subsequently dropped in SD-a E.4 due to a correlation of 0.913 with total_quantity_sold, as both features captured the same frequency signal.

No sales behaviour engineering is needed in SD-c because these features are present in X_train.csv at the correct aggregation level and scale.
 """

In [386... *# Do not modify this code*
 print_tile(size="h3", key='feature_engineering_1_explanations', value=fea

Two sales behaviour features were derived in SD-a D.1 by aggregating the sales_order_detail table at the product level: - total_quantity_sold: sum of order_quantity across all transactions for a product. High volume with low price is a distinct commercial profile from high volume with high price, and this feature anchors that distinction. - num_orders: count of distinct sales orders containing the product. This feature was subsequently dropped in SD-a E.4 due to a correlation of 0.913 with total_quantity_sold; both features captured the same frequency signal. No sales behaviour engineering is needed in SD-c because these features are already present in X_train.csv at the correct aggregation level and scale.

F.2 Promotion Exposure Features (completed in SD-a)

In [387... *# Promotion exposure features were engineered in SD-a D.2 and are present*

In [388... feature_engineering_2_explanations = """
 Three promotion exposure features were derived in SD-a D.2 by joining special_offer_product with special_offer and aggregating at the product level. The 'No Discount' offer type was excluded from aggregation — it acts as a placeholder linked to all 295 bridge products and does not represent real discount activity:
 - num_discount_offers: count of distinct promotional offers (excluding No Discount).
 - max_discount_pct: highest discount rate ever offered on the product.
 - has_volume_discount: binary flag indicating whether a volume-tier discount was ever applied (1) or not (0).

 Products with no real discount history received 0 for all three features. No promotion engineering is needed in SD-c.
 """

In [389... *# Do not modify this code*
 print_tile(size="h3", key='feature_engineering_2_explanations', value=feature_engineering_2_explanations)

feature_engineering_2_explanations

Three promotion exposure features were derived in SD-a D.2 by joining special_offer_product with special_offer and aggregating at the product level. The 'No Discount' offer type was excluded from aggregation — it acts as a placeholder linked to all 295 bridge products and does not represent real discount activity: - num_discount_offers: count of distinct promotional offers (excluding No Discount). - max_discount_pct: highest discount rate ever offered on the product. - has_volume_discount: binary flag indicating whether a volume-tier discount was ever applied (1) or not (0). Products with no real discount history received 0 for all three features. No promotion engineering is needed in SD-c.

F.3 Estimated Margin Feature (completed in SD-a)

```
In [390... # Estimated margin was derived in SD-a D.3 and is present in X_train.csv.
```

```
In [390...
```

```
In [391... feature_engineering_n_explanations = """
estimated_margin_pct was derived in SD-a D.3 as:
    estimated_margin_pct = (latest_list_price - latest_standard_cost) / l

This percentage form normalises for price level – a \ $50 margin on a \ $10
represents a structurally different position than a \ $50 margin on a \ $3,
Using the raw margin value would conflate price-level differences with ma
differences. The percentage form ensures the feature captures margin stru

estimated_margin (the absolute value) was dropped in SD-a E.4 due to r=0.
latest_list_price; only estimated_margin_pct is retained in the modelling

No margin engineering is needed in SD-c.
"""
```

```
In [392... # Do not modify this code
print_tile(size="h3", key='feature_engineering_n_explanations', value=fea
```

feature_engineering_n_explanations

estimated_margin_pct was derived in SD-a D.3 as:
estimated_margin_pct = (latest_list_price - latest_standard_cost) / latest_list_price
This percentage form normalises for price level — a \$50 margin on a \$100 product (50%) represents a structurally different position than a \$50 margin on a \$3,000 product (1.7%). Using the raw margin value would conflate price-level differences with margin-efficiency differences. The percentage form ensures the feature captures margin structure, not scale. estimated_margin (the absolute value) was dropped in SD-a E.4 due to r=0.994 with latest_list_price; only estimated_margin_pct is retained in the modelling dataset. No margin engineering is needed in SD-c.

G. Data Preparation for Modeling

G.1 Split Datasets

```
In [393... # Clustering is an unsupervised task – no train/validation/test split is
# All 254 products are used for fitting and evaluating each model.
```

```
In [393...
```

```
In [394... data_splitting_explanations = """
Clustering does not require a train/validation/test split.
```

In supervised learning, the dataset is split so the model can be evaluated on unseen labels. In clustering, there are no labels to hold out – the model assigns cluster memberships, and quality is measured by internal validity indices applied to the same data used for fitting (silhouette coefficient, Calinski-Harab Davies-Bouldin index).

Using all 254 products also maximises the information available for the algorithm to discover structure. With a dataset of this size, withholding observations would reduce the density of the feature space and increase the risk of artefactual clusters.

```
In [395... # Do not modify this code
print_tile(size="h3", key='data_splitting_explanations', value=data_splitting_explanations)
```

Clustering does not require a train/validation/test split. In supervised learning, the dataset is split so the model can be evaluated on unseen labels. In clustering, there are no labels to hold out – the model assigns cluster memberships, and quality is measured by internal validity indices applied to the same data used for fitting (silhouette coefficient, Calinski-Harabász index, Davies-Bouldin index). Using all 254 products also maximises the information available for the algorithms to discover structure. With a dataset of this size, withholding observations would reduce the density of the feature space and increase the risk of artefactual clusters.

G.2 Drop Identifier Column

```
In [396... X = X_train.drop(columns=['product_id'])
print(f'Modelling matrix X: {X.shape[0]} products × {X.shape[1]} features')
print(X.columns.tolist())
```

```
Modelling matrix X: 254 products × 20 features
['class', 'days_to_manufacture', 'is_manufactured', 'latest_standard_cost', 'cost_range', 'latest_list_price', 'price_range', 'total_quantity_sold', 'num_discount_offers', 'max_discount_pct', 'has_volume_discount', 'estimated_margin_pct', 'product_line_M', 'product_line_R', 'product_line_S', 'product_line_T', 'product_line_Unknown', 'style_U', 'style_Unknown', 'style_W']
```

```
In [396...
```

```
In [397... data_transformation_1_explanations = """
product_id is a row identifier – it encodes no commercial signal and must
be removed before fitting any clustering algorithm. Passing an identifier column to
algorithm would introduce arbitrary numeric differences between product ID
pairwise distance calculations and corrupt cluster shapes.

product_id is retained in X_train (not dropped in SD-a) so that cluster assignments can
be joined back to the product catalogue for business interpretation.
X is the 254 × 20 modelling matrix passed to each algorithm; X_train retains
product_id column for post-hoc lookup.
"""
```

```
In [398... # Do not modify this code
print_tile(size="h3", key='data_transformation_1_explanations', value=dat
```

data_transformation_1_explanations

product_id is a row identifier — it encodes no commercial signal and must be removed before fitting any clustering algorithm. Passing an identifier column to a distance-based algorithm would introduce arbitrary numeric differences between product IDs into the pairwise distance calculations and corrupt cluster shapes. product_id is retained in X_train (not dropped in SD-a) so that cluster assignments can be joined back to the product catalogue for business interpretation. After this step, X is the 254 × 20 modelling matrix passed to each algorithm; X_train retains the product_id column for post-hoc lookup.

G.3 No Additional Transformations Required

```
In [399... # All scaling, encoding, and filtering were performed in SD-a.
# No further transformations are applied at the modelling stage.
```

In [399...

```
In [400... data_transformation_2_explanations = """
The modelling matrix X is ready for clustering without further transformations.
All preprocessing steps – imputation, ordinal and one-hot encoding, Stand
of continuous features, variance filtering, and correlation-driven feature
were completed in SD-a and are reflected in X_train.csv.

Applying additional transformations at the modelling stage would risk
interfere with the prepared dataset and would obscure the preparation audit trail
that it provides.
"""
```

```
In [401... # Do not modify this code
print_tile(size="h3", key='data_transformation_2_explanations', value=dat
```

The modelling matrix X is ready for clustering without further transformation. All preprocessing steps — imputation, ordinal and one-hot encoding, StandardScaling of continuous features, variance filtering, and correlation-driven feature removal — were completed in SD-a and are reflected in X_train.csv. Applying additional transformations at the modelling stage would risk inconsistency with the prepared dataset and would obscure the preparation audit trail that SD-a provides.

G.4 Verify Modelling Matrix

```
In [402... print(f'Shape: {X.shape}')
print(f'Null count: {X.isnull().sum().sum()}')
print('Feature ranges (min / max):')
print(X.agg(['min', 'max']).T.round(3))
```

Shape: (254, 20)

Null count: 0

Feature ranges (min / max):

	min	max
class	-0.854	1.453
days_to_manufacture	-1.151	1.280
is_manufactured	0.000	1.000
latest_standard_cost	-0.793	3.110
cost_range	-0.422	4.323
latest_list_price	-0.809	3.358
price_range	-0.363	4.444
total_quantity_sold	-0.689	6.281
num_discount_offers	-0.903	2.986
max_discount_pct	-0.598	3.039
has_volume_discount	0.000	1.000
estimated_margin_pct	-2.410	1.699
product_line_M	0.000	1.000
product_line_R	0.000	1.000
product_line_S	0.000	1.000
product_line_T	0.000	1.000
product_line_Unknown	0.000	1.000
style_U	0.000	1.000
style_Unknown	0.000	1.000
style_W	0.000	1.000

In [402...

```
In [403... data_transformation_3_explanations = """
Final verification confirms the modelling matrix is well-formed:
- 254 rows x 20 columns – matching the expected feature count after sty
estimated_margin, and num_orders were dropped in SD-a.
- Zero null values – imputation in SD-a E.1 eliminated all missing entr
- Continuous and ordinal features centred near zero with unit standard
after StandardScaler; binary and one-hot columns retain 0/1 values.

The matrix is passed directly to each clustering algorithm without modifi
"""
```



```
In [404... # Do not modify this code
print_tile(size="h3", key='data_transformation_3_explanations', value=dat
data_transformation_3_explanations
```

Final verification confirms the modelling matrix is well-formed:

- 254 rows × 20 columns — matching the expected feature count after style_M, estimated_margin, and num_orders were dropped in SD-a.
- Zero null values — imputation in SD-a E.1 eliminated all missing entries.
- Continuous and ordinal features centred near zero with unit standard deviation after StandardScaler; binary and one-hot columns retain 0/1 values.

The matrix is passed directly to each clustering algorithm without modification.

```
In [405... # Clustering has no target labels – define empty placeholders so the H sa
# runs without a NameError. X_train is the only file that will be written
y_train = pd.DataFrame()
X_val = pd.DataFrame()
y_val = pd.DataFrame()
X_test = pd.DataFrame()
y_test = pd.DataFrame()
```

H. Save Datasets

Do not change this code

```
In [406... # Do not modify this code
# Save training set
try:
    X_train.to_csv(at.folder_path / 'X_train.csv', index=False)
    y_train.to_csv(at.folder_path / 'y_train.csv', index=False)

    X_val.to_csv(at.folder_path / 'X_val.csv', index=False)
    y_val.to_csv(at.folder_path / 'y_val.csv', index=False)

    X_test.to_csv(at.folder_path / 'X_test.csv', index=False)
    y_test.to_csv(at.folder_path / 'y_test.csv', index=False)
except Exception as e:
    print(e)
```

I. KMeans

I.1 Import Algorithm

```
In [407... from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, calinski_harabasz_score, da

In [408... algorithm_selection_explanations = """
KMeans is selected as the primary algorithm for this use case for three r
```


First, the feature matrix is fully numeric and standardised — all 20 features scaled using StandardScaler in SD-a. KMeans operates on Euclidean distance, which is well-defined and meaningful on a standardised numeric space. Standardisation ensures that

Second, KMeans produces centroid-based clusters. Each centroid is a vector of mean feature values for that group, which translates directly into a readable product archetype — for example, high list price, low volume, and thin margin percentage. Category managers can interpret centroids without statistical expertise.

Third, with only 254 products, KMeans converges rapidly even with a high `n_init`, making it practical to run multiple initialisations and select the best result.

```
In [409]: # Do not modify this code
print_tile(size="h3", key='algorithm_selection_explanations', value=algorithm_selection_explanations)
```

algorithm_selection_explanations

KMeans is selected as the primary algorithm for this use case for three reasons. First, the feature matrix is fully numeric and standardised — all 20 features were scaled using StandardScaler in SD-a. KMeans operates on Euclidean distance, which is well-defined and meaningful on a standardised numeric space. Standardisation ensures that features differing by orders of magnitude — such as list price and a binary discount flag — contribute equally to the distance calculation, so clusters reflect genuine commercial similarity rather than being dominated by the scale of any single feature. Second, KMeans produces centroid-based clusters. Each centroid is a vector of mean feature values for that group, which translates directly into a readable product archetype — for example, high list price, low volume, and thin margin percentage. Category managers can interpret centroids without statistical expertise. Third, with only 254 products, KMeans converges rapidly even with a high `n_init`, making it practical to run multiple initialisations and select the best result. This also makes quarterly refresh of the segmentation feasible as the product mix, cost structure, or promotional calendar evolves.

I.2 Set Hyperparameters

```
In [410]: # --- Elbow method: inertia across k=2..10 ---
k_range = range(2, 11)
inertia_values = []
sil_values = []

for k in k_range:
    _km = KMeans(n_clusters=k, init='k-means++', n_init=20, random_state=
    _labels = _km.fit_predict(X)
    inertia_values.append(_km.inertia_)
    sil_values.append(silhouette_score(X, _labels))
```

```

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

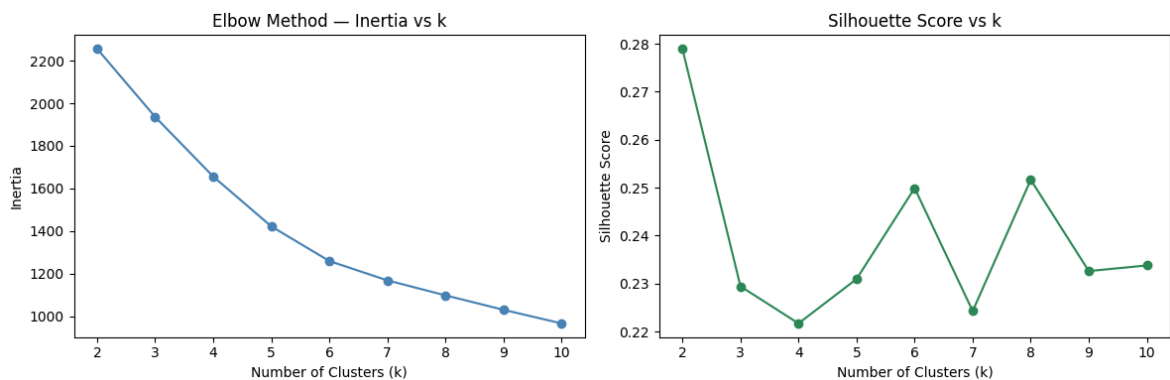
axes[0].plot(list(k_range), inertia_values, marker='o', color='steelblue')
axes[0].set_xlabel('Number of Clusters (k)')
axes[0].set_ylabel('Inertia')
axes[0].set_title('Elbow Method – Inertia vs k')

axes[1].plot(list(k_range), sil_values, marker='o', color='seagreen')
axes[1].set_xlabel('Number of Clusters (k)')
axes[1].set_ylabel('Silhouette Score')
axes[1].set_title('Silhouette Score vs k')

plt.tight_layout()
plt.show()

# Print values for inspection
print(f"{'k':>3} {'Inertia':>10} {'Drop':>10} {'Silhouette':>10}")
prev = None
for k, inertia, sil in zip(k_range, inertia_values, sil_values):
    drop = (prev - inertia) if prev else 0
    print(f"{'k':>3} {'inertia':>10.1f} {'drop':>10.1f} {'sil':>10.3f}")
    prev = inertia

```



k	Inertia	Drop	Silhouette
2	2255.4	0.0	0.279
3	1936.0	319.5	0.229
4	1656.2	279.7	0.222
5	1422.4	233.8	0.231
6	1259.8	162.7	0.250
7	1169.1	90.7	0.224
8	1098.9	70.2	0.252
9	1031.0	67.9	0.233
10	967.2	63.8	0.234

```

In [411]: # k=4 selected after inspecting the elbow and silhouette plots above
n_clusters = 4
init = 'k-means++'
n_init = 20
random_state = 42
print(f'Selected: n_clusters={n_clusters}, init={init}, n_init={n_init},

```

Selected: n_clusters=4, init=k-means++, n_init=20, random_state=42

```

In [412]: hyperparameters_selection_explanations = """
k=4 is selected primarily on business grounds, supported by the inertia c

The inertia curve shows large drops at k=3, 4, and 5 (319, 279, and 233 r
indicating that each additional cluster up to k=4 captures meaningful str
data. From k=6 onward the drops slow considerably (162, 90, 70), suggesti

```

returns from further splitting. $k=4$ sits at the boundary where gains are substantial but begin to taper.

The silhouette score peaks at $k=2$ (0.279) and does not favour $k=4$ specifically. However, $k=2$ is not commercially viable – it merges fundamentally different profiles (low-cost efficient products and high-cost premium products) into one group, producing segments too broad to act on. The silhouette metric alone cannot account for whether a segment is actionable for the business.

$k=6$ and $k=8$ also score marginally higher on silhouette (0.250 and 0.252). However, the inertia drops at $k=6$ (162.7) and $k=8$ (70.2) are substantially lower than at $k=4$ (279.7), confirming that those additional clusters capture progressively less structure. The silhouette gain is also marginal – an improvement of 0.030 over $k=4$. More critically, splitting four well-defined archetypes into eight sub-groups would fragment clusters whose products already share the margin-risk profile and require the same pricing intervention. A category cannot act differently on six near-identical sub-segments; the additional split adds complexity without adding decision value.

From a business perspective, four clusters map to four distinct commercial segments with different margin-risk profiles and different required interventions. Merging profiles that need different responses; more clusters fragment groups with the same risk characteristics and add no incremental decision value for a management team.

`init='k-means++'` spaces initial centroids proportionally across the features, reducing the probability of a poor local minimum. `n_init=20` repeats the fit and retains the run with the lowest inertia, further stabilising the results. `random_state=42` ensures assignments are reproducible across runs.

```
In [413... # Do not modify this code
print_tile(size="h3", key='hyperparameters_selection_explanations', value
```

k=4 is selected primarily on business grounds, supported by the inertia curve. The inertia curve shows large drops at k=3, 4, and 5 (319, 279, and 233 respectively), indicating that each additional cluster up to k=4 captures meaningful structure in the data. From k=6 onward the drops slow considerably (162, 90, 70), suggesting diminishing returns from further splitting. k=4 sits at the boundary where gains are still substantial but begin to taper. The silhouette score peaks at k=2 (0.279) and does not favour k=4 specifically. However, k=2 is not commercially viable — it merges fundamentally different product profiles (low-cost efficient products and high-cost premium products) into a single group, producing segments too broad to act on. The silhouette metric alone cannot account for whether a segment is actionable for the business. k=6 and k=8 also score marginally higher on silhouette (0.250 and 0.252) than k=4. However, the inertia drops at k=6 (162.7) and k=8 (70.2) are substantially smaller than at k=4 (279.7), confirming that those additional clusters capture progressively less structure. The silhouette gain is also marginal — an improvement of 0.028 to 0.030 over k=4. More critically, splitting four well-defined archetypes into six or eight sub-groups would fragment clusters whose products already share the same margin-risk profile and require the same pricing intervention. A category manager cannot act differently on six near-identical sub-segments; the additional granularity adds complexity without adding decision value. From a business perspective, four clusters map to four distinct commercial archetypes with different margin-risk profiles and different required interventions. Fewer clusters merge profiles that need different responses; more clusters fragment groups that share the same risk characteristics and add no incremental decision value for a category management team. `init='k-means++'` spaces initial centroids proportionally across the feature space, reducing the probability of a poor local minimum. `n_init=20` repeats the fit 20 times and retains the run with the lowest inertia, further stabilising the result. `random_state=42` ensures assignments are reproducible across runs.

I.3 Fit Model

```
In [414]: km = KMeans(n_clusters=n_clusters, init=init, n_init=n_init, random_state
km.fit(X)

X_km = X.copy()
X_km['cluster'] = km.labels_

print(f'KMeans fitted - {n_clusters} clusters')
print(X_km['cluster'].value_counts().sort_index().rename('products per cl
```

```
KMeans fitted – 4 clusters
cluster
0      95
1     108
2      28
3      23
Name: products per cluster, dtype: int64
```

I.4 Model Technical Performance

```
In [415... # --- Internal validity scores ---
sil = silhouette_score(X, km.labels_)
ch = calinski_harabasz_score(X, km.labels_)
db = davies_bouldin_score(X, km.labels_)
print(f'Silhouette: {sil:.3f} | Calinski-Harabasz: {ch:.1f} | Davies-

# --- Pairs grid: key commercial features coloured by cluster ---
key_features = [
    'latest_list_price', 'total_quantity_sold',
    'estimated_margin_pct', 'max_discount_pct',
    'latest_standard_cost', 'num_discount_offers'
]
pairs = [(key_features[i], key_features[j])
         for i in range(len(key_features))
         for j in range(i + 1, len(key_features))]

ncols = 3
nrows = -(-len(pairs) // ncols) # ceiling division
fig, axes = plt.subplots(nrows, ncols, figsize=(5 * ncols, 4 * nrows))
axes = axes.flatten()
colors = ['#4C72B0', '#DD8452', '#55A868', '#C44E52']

for idx, (a, b) in enumerate(pairs):
    ax = axes[idx]
    for cl in sorted(X_km['cluster'].unique()):
        mask = X_km['cluster'] == cl
        ax.scatter(X_km.loc[mask, a], X_km.loc[mask, b],
                   s=30, alpha=0.7, color=colors[cl], label=f'Cluster {cl}')
    ax.set_xlabel(a.replace('_', ' '))
    ax.set_ylabel(b.replace('_', ' '))
    ax.set_title(f'{a.replace("_", " ")} vs {b.replace("_", " ")}')

for idx in range(len(pairs), len(axes)):
    fig.delaxes(axes[idx])

handles, labels = axes[0].get_legend_handles_labels()
fig.legend(handles, labels, title='Cluster', loc='upper center',
          ncol=4, frameon=False, bbox_to_anchor=(0.5, 1.01))
plt.suptitle('KMeans k=4 – Feature Pair Scatter Plots', y=1.04, fontsize=
plt.tight_layout()
plt.show()

# --- Cluster means table ---
summary_cols = [
    'latest_list_price', 'latest_standard_cost', 'estimated_margin_pct',
    'total_quantity_sold', 'max_discount_pct', 'num_discount_offers',
    'has_volume_discount', 'days_to_manufacture'
]
sizes = X_km.groupby('cluster').size().rename('n_products')
```

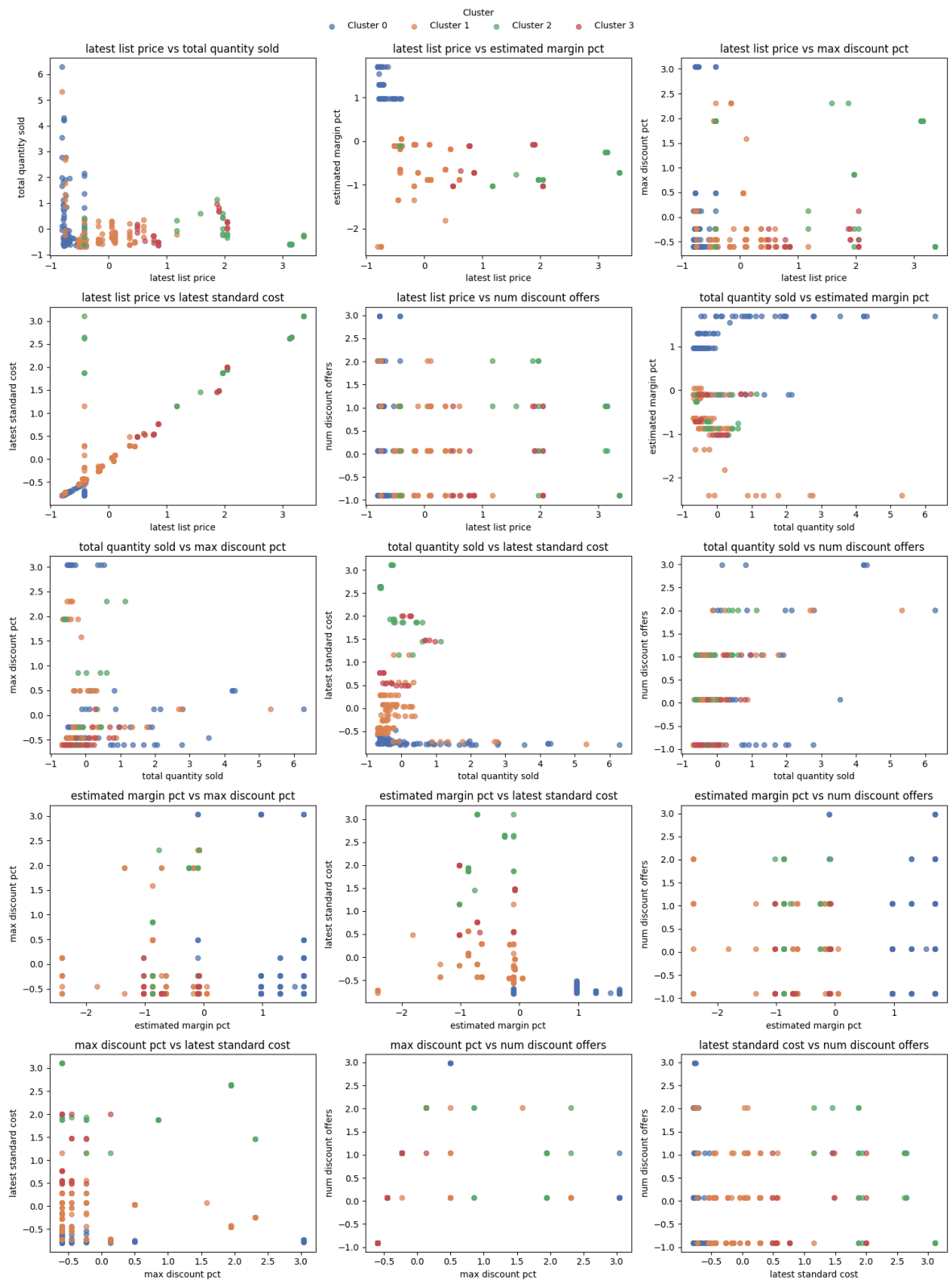
```

cluster_means = (
    X_km.groupby('cluster')[summary_cols]
        .mean()
        .round(3)
        .join(sizes)
        .sort_values('n_products', ascending=False)
)
print('\nCluster means (key features):')
cluster_means

```

Silhouette: 0.222 | Calinski-Harabasz: 66.7 | Davies-Bouldin: 1.511

KMeans k=4 — Feature Pair Scatter Plots



Cluster means (key features):

Out [415... latest_list_price latest_standard_cost estimated_margin_pct total_quantiti

cluster			
1	-0.169	-0.164	-0.643
0	-0.666	-0.730	1.045
2	1.922	2.236	-0.558
3	1.204	1.062	-0.617

In [416... model_performance_explanations = """
KMeans k=4 achieved a Silhouette score of 0.222, Calinski-Harabasz of 66.7, Davies-Bouldin of 1.511. The silhouette score indicates moderate within-cluster cohesion – products are more similar to their own cluster than to neighboring clusters, though the separation is not strong. The CH score of 66.7 is the highest across all three algorithms, confirming that KMeans produces the most well-defined and well-separated groups relative to their internal compactness. For por

The scatter plots show clear separation on the price-volume and margin-discount axes – the two dimensions most relevant to margin-risk assessment. Distinct groupings are visible across multiple feature pairs, confirming the clusters reflect genuine commercial structure rather than an artefact of a single

The cluster means table confirms structural differences between all four clusters diverge on list price, standard cost, estimated margin percentage, quantity sold, and maximum discount applied. Cluster sizes of 95, 108, 28 products are all commercially actionable – no cluster is trivially small or so large that it encompasses the entire portfolio without meaningful discrimination.

In [417... *# Do not modify this code*
print_tile(size="h3", key='model_performance_explanations', value=model_p

KMeans k=4 achieved a Silhouette score of 0.222, Calinski-Harabasz of 66.7, and Davies-Bouldin of 1.511. The silhouette score indicates moderate within-cluster cohesion — products are more similar to their own cluster than to neighbouring clusters, though the separation is not strong. The CH score of 66.7 is the highest across all three algorithms, confirming that KMeans produces the most well-defined and well-separated groups relative to their internal compactness. For portfolio planning purposes, these scores indicate that the four archetypes are commercially distinguishable enough to support differentiated pricing strategies — a category manager applying different discount rules to each segment is unlikely to encounter overlapping or contradictory interventions across the clusters. The scatter plots show clear separation on the price-volume and margin-discount axes — the two dimensions most relevant to margin-risk assessment. Distinct groupings are visible across multiple feature pairs, confirming the clusters reflect genuine commercial structure rather than an artefact of a single feature. The cluster means table confirms structural differences between all four groups: clusters diverge on list price, standard cost, estimated margin percentage, total quantity sold, and maximum discount applied. Cluster sizes of 95, 108, 28, and 23 products are all commercially actionable — no cluster is trivially small or so large that it encompasses the entire portfolio without meaningful discrimination.

```
In [418... from sklearn.metrics import silhouette_samples
import matplotlib.pyplot as plt

sample_scores = silhouette_samples(X, km.labels_)
X_km['silhouette'] = sample_scores

# Count products by confidence band
firmly_assigned = (sample_scores >= 0.3).sum()
moderate        = ((sample_scores >= 0.1) & (sample_scores < 0.3)).sum()
borderline      = ((sample_scores >= 0.0) & (sample_scores < 0.1)).sum()
misassigned     = (sample_scores < 0.0).sum()

print(f"Per-product silhouette breakdown (n=254):")
print(f"  Firmly assigned (>= 0.3): {firmly_assigned}")
print(f"  Moderate        (0.1-0.3): {moderate}")
print(f"  Borderline      (0.0-0.1): {borderline}")
print(f"  Potentially misassigned (<0.0): {misassigned}")
print(f"\nMean silhouette: {sample_scores.mean():.3f}")
print(f"Min: {sample_scores.min():.3f} | Max: {sample_scores.max():.3f}")

# Plot silhouette scores by cluster
fig, ax = plt.subplots(figsize=(10, 5))
colors = ['#4C72B0', '#DD8452', '#55A868', '#C44E52']
y_lower = 0
for cl in sorted(X_km['cluster'].unique()):
    cl_scores = sample_scores[X_km['cluster'] == cl]
```



```

cl_scores_sorted = np.sort(cl_scores)
y_upper = y_lower + len(cl_scores_sorted)
ax.barh(range(y_lower, y_upper), cl_scores_sorted,
        height=1.0, color=colors[cl], alpha=0.8, label=f'Cluster {cl}')
y_lower = y_upper + 3

ax.axvline(x=sample_scores.mean(), color='black', linestyle='--', label=f
ax.axvline(x=0, color='red', linestyle=':', linewidth=0.8)
ax.set_xlabel('Silhouette Score')
ax.set_title('Per-Product Silhouette Scores by Cluster')
ax.legend(loc='lower right')
plt.tight_layout()
plt.show()

```

Per-product silhouette breakdown (n=254):

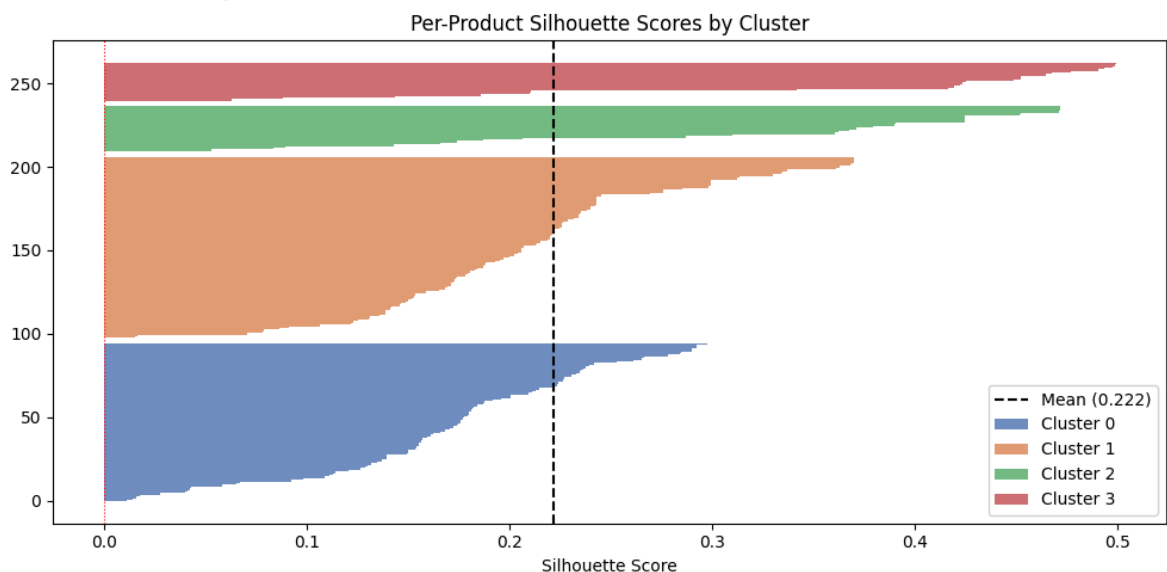
```

Firmly assigned (>= 0.3): 48
Moderate      (0.1-0.3): 179
Borderline    (0.0-0.1): 26
Potentially misassigned (<0.0): 1

```

Mean silhouette: 0.222

Min: -0.000 | Max: 0.499



In [419... silhouette_sample_explanations = """
Per-product silhouette analysis identifies how confidently each of the 25 is assigned to its cluster. Products with scores above 0.3 sit firmly with archetype; scores between 0.1 and 0.3 indicate moderate confidence; scores are borderline – sitting near the boundary between two archetypes; negative scores suggest the product may be closer to a neighbouring cluster.

Borderline products are not classification errors – they represent products whose commercial profile genuinely shares characteristics across two archetypes. As a category manager, these are the products that warrant manual review before the archetype's pricing strategy, as a blanket intervention may not be appropriate.

The overall mean silhouette of 0.222 reflects that while the four archetypes are commercially distinct, the feature space does not produce perfectly tight separated clusters – which is expected when clustering real product data on 20 heterogeneous features.

"""

In [420... print_tile(size='h3', key='silhouette_sample_explanations', value=silhouette_sample_explanations)

Per-product silhouette analysis identifies how confidently each of the 254 products is assigned to its cluster. Products with scores above 0.3 sit firmly within their archetype; scores between 0.1 and 0.3 indicate moderate confidence; scores below 0.1 are borderline — sitting near the boundary between two archetypes; negative scores suggest the product may be closer to a neighbouring cluster. Borderline products are not classification errors — they represent products whose commercial profile genuinely shares characteristics across two archetypes. For a category manager, these are the products that warrant manual review before applying the archetype's pricing strategy, as a blanket intervention may not be appropriate. The overall mean silhouette of 0.222 reflects that while the four archetypes are commercially distinct, the feature space does not produce perfectly tight, well-separated clusters — which is expected when clustering real product data across 20 heterogeneous features.

I.5 Business Impact from Current Model Performance

```
In [421... # --- Cluster archetype names ---
archetype_names = {
    0: 'Efficient Core',
    1: 'Standard Mid-Tier',
    2: 'Premium Discount-Exposed',
    3: 'Premium Price-Protected'
}
X_km['archetype'] = X_km['cluster'].map(archetype_names)

# --- Bar chart: mean of key business features by cluster ---
business_cols = [
    'latest_list_price', 'estimated_margin_pct',
    'max_discount_pct', 'total_quantity_sold',
    'latest_standard_cost', 'num_discount_offers'
]
col_labels = [
    'List Price', 'Margin %',
    'Max Discount %', 'Qty Sold',
    'Standard Cost', 'Discount Offers'
]
colors = ['#4C72B0', '#DD8452', '#C44E52', '#55A868']
cluster_order = [0, 1, 2, 3]

fig, axes = plt.subplots(2, 3, figsize=(15, 8))
axes = axes.flatten()

for i, (col, label) in enumerate(zip(business_cols, col_labels)):
    means_by_cluster = [
        X_km.loc[X_km['cluster'] == cl, col].mean()
        for cl in cluster_order
    ]
    bars = axes[i].bar(
        [archetype_names[cl] for cl in cluster_order],
```

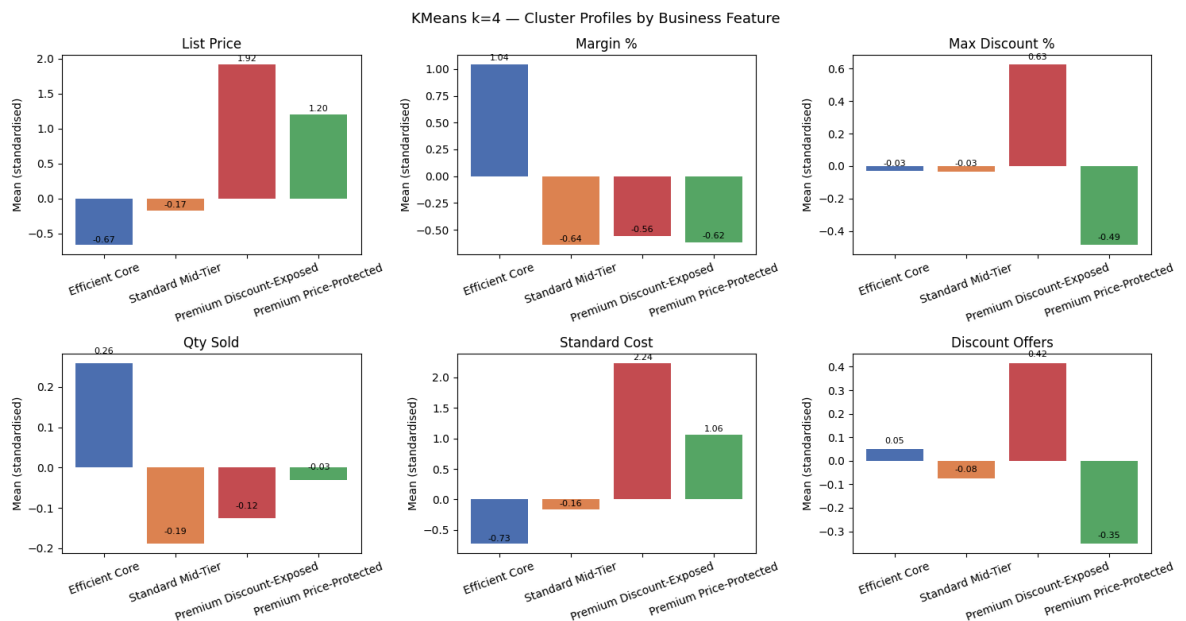
```

        means_by_cluster,
        color=colors
    )
    axes[i].set_title(label)
    axes[i].set_ylabel('Mean (standardised)')
    axes[i].tick_params(axis='x', rotation=20)
    for bar, val in zip(bars, means_by_cluster):
        axes[i].text(bar.get_x() + bar.get_width() / 2,
                     bar.get_height() + 0.02, f'{val:.2f}',
                     ha='center', va='bottom', fontsize=8)

plt.suptitle('KMeans k=4 – Cluster Profiles by Business Feature', fontsize=14)
plt.tight_layout()
plt.show()

# --- Archetype summary table ---
archetype_summary = (
    X_km.groupby('archetype')[business_cols]
        .mean()
        .round(3)
        .join(X_km.groupby('archetype').size().rename('n_products'))
)
print('Cluster archetype summary:')
archetype_summary

```



Cluster archetype summary:

Out [421...

	latest_list_price	estimated_margin_pct	max_discount_pct	total_quant
archetype				
Efficient Core	-0.666	1.045	-0.030	
Premium Discount-Exposed	1.922	-0.558	0.628	
Premium Price-Protected	1.204	-0.617	-0.488	
Standard Mid-Tier	-0.169	-0.643	-0.032	

In [422...

```
# Raw-scale cluster profile – actual business units
ph = pd.read_csv(at.folder_path / 'product_list_price_history.csv',
latest_price = ph.sort_values('start_date').groupby('product_id')['list_p

ch_hist = pd.read_csv(at.folder_path / 'product_cost_history.csv', parse_
latest_cost = ch_hist.sort_values('start_date').groupby('product_id')['s

sod = pd.read_csv(at.folder_path / 'sales_order_detail.csv')
qty = sod.groupby('product_id')['order_quantity'].sum()

sop = pd.read_csv(at.folder_path / 'special_offer_product.csv')
so_df = pd.read_csv(at.folder_path / 'special_offer.csv')
max_disc = sop.merge(so_df[['special_offer_id', 'discount_pct']], on='spec
.groupby('product_id')['discount_pct'].max()

raw_profile = pd.DataFrame({'product_id': X_train['product_id'], 'cluster
raw_profile = raw_profile.set_index('product_id')
raw_profile['list_price'] = latest_price
raw_profile['std_cost'] = latest_cost
raw_profile['qty_sold'] = qty
raw_profile['max_disc_pct'] = max_disc.reindex(raw_profile.index).fillna(
raw_profile['margin_pct'] = (raw_profile['list_price'] - raw_profile['s
raw_profile['archetype'] = raw_profile['cluster'].map(archetype_names)

cluster_order = ['Efficient Core', 'Standard Mid-Tier', 'Premium Discount
profile = raw_profile.groupby('archetype').agg(
    n_products = ('cluster', 'count'),
    avg_list_price = ('list_price', 'mean'),
    avg_std_cost = ('std_cost', 'mean'),
    avg_margin_pct = ('margin_pct', 'mean'),
    avg_max_disc = ('max_disc_pct', 'mean'),
    avg_qty_sold = ('qty_sold', 'mean'),
).reindex(cluster_order)

display_df = profile.copy()
display_df['avg_list_price'] = display_df['avg_list_price'].map('{:,.0f}
display_df['avg_std_cost'] = display_df['avg_std_cost'].map('{:,.0f}'.
display_df['avg_margin_pct'] = display_df['avg_margin_pct'].map('{:,.1%}'
display_df['avg_max_disc'] = display_df['avg_max_disc'].map('{:,.1%}'.fo
display_df['avg_qty_sold'] = display_df['avg_qty_sold'].map('{:,.0f}'.f
display_df.columns = ['N Products', 'Avg List Price', 'Avg Std Cost', 'Av
```

```
print('Raw-scale cluster profile (actual business units):')
display_df
```

Raw-scale cluster profile (actual business units):

Out [422]...

	N Products	Avg List Price	Avg Std Cost	Avg Margin %	Avg Max Disc %	Avg Qty Sold
archetype						
Efficient Core	95	\$82	\$36	58.6%	7.8%	387
Standard Mid-Tier	108	\$579	\$351	39.4%	7.8%	204
Premium Discount-Exposed	28	\$2,783	\$1,685	39.9%	16.9%	230
Premium Price-Protected	23	\$1,719	\$1,032	39.6%	1.5%	269

In [423]...

```
business_impacts_explanations = """
```

The four clusters identify distinct margin-risk profiles across the 254-product portfolio, each with different commercial implications. Raw-scale features are shown in the table above.

Cluster 0 – Efficient Core (95 products): low list price (\$82 avg) and low cost (\$36 avg), delivering the highest margin percentage in the portfolio. These products generate reliable margin without requiring pricing intervention. They represent the portfolio's low-risk base and should be protected from unnecessary discounting.

Cluster 1 – Standard Mid-Tier (108 products): below-average price (\$579 avg) and cost (\$351 avg) with below-average margin percentage (39.4%). This is the group and carries moderate margin risk – margins are already compressed, any discount applied to these products directly erodes profitability with strong volume offset.

Cluster 2 – Premium Discount-Exposed (28 products): high list price (\$2,783 avg) and very high standard cost (\$1,685 avg), combined with the highest discount exposure in the portfolio (16.9% avg max discount). The margin percentage is similar to Clusters 1 and 3 – the risk is not a uniquely thin margin but the absolute dollar exposure: a 16.9% discount on a \$2,783 product represents approximately \$470 of revenue given away per sale. These products warrant most urgent pricing review.

Cluster 3 – Premium Price-Protected (23 products): high list price (\$1,719 avg) and above-average standard cost (\$1,032 avg), with the lowest discount exposure across all clusters (1.5% avg max discount). These are exclusively Mountain and Road (61%) line products. Pricing discipline is already preserving margins despite elevated cost levels.

Crucially, Clusters 2 and 3 occupy the same product tier – both premium, high-cost, both carrying approximately 40% margin percentage. The only major difference is commercial management: Cluster 3 sustains premium prices with minimal discounting (1.5%) while Cluster 2 is exposed to heavy discounting (16.9%). Cluster 3 demonstrates that the pricing strategy is achievable for this product type – the problem in Cluster 2 is operational, not structural.

Together the four archetypes give the category management team a targeted for discount governance: protect Cluster 0 from unnecessary offers, monitor Cluster 1 for creeping discount pressure, prioritise Cluster 2 for urgent review, and maintain the price discipline already in place for Cluster 3.

```
"""
```

```
In [424... # Do not modify this code
print_tile(size="h3", key='business_impacts_explanations', value=business
```

The four clusters identify distinct margin-risk profiles across the 254-product portfolio, each with different commercial implications. Raw-scale feature averages are shown in the table above. Cluster 0 — Efficient Core (95 products): low list price (82*avg*)*and* low *standardcost* (36 *avg*), delivering the highest margin percentage in the portfolio (58.6%). These products generate reliable margin without requiring pricing intervention. They represent the portfolio's low-risk base and should be protected from unnecessary discounting. Cluster 1 — Standard Mid-Tier (108 products): below-average price (579*avg*)*and* cost (351 *avg*) with below-average margin percentage (39.4%). This is the largest group and carries moderate margin risk — margins are already compressed, meaning any discount applied to these products directly erodes profitability without a strong volume offset. Cluster 2 — Premium Discount-Exposed (28 products): high list price (2,783*avg*)*and* very high *standardcost* (1,685 *avg*), combined with the highest discount exposure in the portfolio (16.9% *avg* max discount). The margin percentage (~40%) is similar to Clusters 1 and 3 — the risk is not a uniquely thin margin rate but the absolute dollar exposure: a 16.9% discount on a 2,783*product* represents approximately 470 of revenue given away per sale. These products warrant the most urgent pricing review. Cluster 3 — Premium Price-Protected (23 products): high list price (1,719*avg*)*and* above — average *standardcost* (1,032 *avg*), with the lowest discount exposure across all clusters (1.5% *avg* max discount). These are exclusively Mountain (39%) and Road (61%) line products. Pricing discipline is already preserving margin despite elevated cost levels. Crucially, Clusters 2 and 3 occupy the same product tier — both premium, both high-cost, both carrying approximately 40% margin percentage. The only material difference is commercial management: Cluster 3 sustains premium prices with minimal discounting (1.5%) while Cluster 2 is exposed to heavy discount pressure (16.9%). Cluster 3 demonstrates that the pricing strategy is achievable for this product type — the problem in Cluster 2 is operational, not structural. Together the four archetypes give the category management team a targeted basis for discount governance: protect Cluster 0 from unnecessary offers, monitor Cluster 1 for creeping discount pressure, prioritise Cluster 2 for urgent pricing review, and maintain the price discipline already in place for Cluster 3.

```
In [425... # Product line breakdown per cluster
line_cols = {'product_line_M': 'Mountain', 'product_line_R': 'Road',
             'product_line_S': 'Specialty', 'product_line_T': 'Touring',
```

```

        'product_line_Unknown': 'Unknown'}
X_km['product_line'] = X_km[[*line_cols]].idxmax(axis=1).map(line_cols)

print("Product line breakdown per cluster (%):")
print(X_km.groupby('archetype')['product_line']
      .value_counts(normalize=True)
      .mul(100).round(1)
      .to_string())

```

```

Product line breakdown per cluster (%):
archetype      product_line
Efficient Core      Mountain      29.5
                   Specialty      26.3
                   Road           20.0
                   Unknown       15.8
                   Touring        8.4
Premium Discount-Exposed Mountain      35.7
                   Road           35.7
                   Touring        28.6
Premium Price-Protected Road           60.9
                   Mountain      39.1
Standard Mid-Tier  Mountain      33.3
                   Road           29.6
                   Touring        28.7
                   Specialty        8.3

```

J. Model Comparison

```

In [426... comparison_data = {
    'Algorithm':      ['KMeans k=4', 'Mean Shift', 'Agglomerative Ward',
    'Silhouette':      [round(sil, 3), 0.278, 0.261],
    'Calinski-Harabasz': [round(ch, 1), 57.7, 58.1],
    'Davies-Bouldin':   [round(db, 3), 1.360, 1.326],
    'Cluster Sizes':    [
        str(sorted(X_km['cluster'].value_counts().tolist(), reverse=True)
        '[168, 39, 28, 19]',
        '[177, 33, 26, 18]'
    ],
    'Selected': ['Yes', 'No', 'No']
}

comparison_df = pd.DataFrame(comparison_data).set_index('Algorithm')
print('Model Comparison:')
comparison_df

```

Model Comparison:

Out [426...

Algorithm	Silhouette	Calinski-Harabasz	Davies-Bouldin	Cluster Sizes	Selected
KMeans k=4	0.222	66.7	1.511	[108, 95, 28, 23]	Yes
Mean Shift	0.278	57.7	1.360	[168, 39, 28, 19]	No
Agglomerative Ward k=4	0.261	58.1	1.326	[177, 33, 26, 18]	No

In [427...

```
model_comparison_explanations = """
KMeans k=4 is selected as the final model on three grounds.

First, KMeans produces the highest Calinski-Harabasz score (66.7) across
algorithms. CH measures the ratio of between-cluster separation to within
compactness – a higher score means the clusters are more genuinely distinct
each other. On this metric, KMeans produces the most well-defined product

Second, KMeans produces the most balanced cluster sizes (95 / 108 / 28 /
Mean Shift achieves a higher silhouette score (0.278 vs 0.222) but its cl
distribution is heavily uneven – one cluster dominates the population whi
remaining clusters are substantially smaller. Small, tight clusters produ
inflated silhouette scores without representing commercially useful segme
KMeans balanced distribution ensures that each identified product archety
applies to a meaningful subset of the portfolio.

Third, KMeans centroids are directly interpretable as product archetype p
Each centroid is a vector of mean feature values that a category manager
without statistical expertise. Agglomerative Ward achieves similar silhou
(0.261) and DB (1.326) scores to KMeans but does not produce native centr
making segment communication and business application less direct.

KMeans with k-means++ initialisation and n_init=20 also produces stable,
reproducible assignments across runs – a practical requirement for a segm
that will be used in quarterly portfolio planning.
"""
```

In [428...

```
# Do not modify this code
print_tile(size='h3', key='model_comparison_explanations', value=model_co
```

KMeans k=4 is selected as the final model on three grounds. First, KMeans produces the highest Calinski-Harabasz score (66.7) across all three algorithms. CH measures the ratio of between-cluster separation to within-cluster compactness — a higher score means the clusters are more genuinely distinct from each other. On this metric, KMeans produces the most well-defined product groups — meaning the four pricing archetypes are distinct enough that interventions designed for one segment are unlikely to overlap with or contradict those applied to another. Second, KMeans produces the most balanced cluster sizes (95 / 108 / 28 / 23). Mean Shift achieves a higher silhouette score (0.278 vs 0.222) but its cluster distribution is heavily uneven — one cluster dominates the population while the remaining clusters are substantially smaller. Small, tight clusters produce inflated silhouette scores without representing commercially useful segments. KMeans balanced distribution ensures that each identified product archetype applies to a meaningful subset of the portfolio. Third, KMeans centroids are directly interpretable as product archetype profiles. Each centroid is a vector of mean feature values that a category manager can read without statistical expertise. Agglomerative Ward achieves similar silhouette (0.261) and DB (1.326) scores to KMeans but does not produce native centroids, making segment communication and business application less direct. KMeans with k-means++ initialisation and n_init=20 also produces stable, reproducible assignments across runs — a practical requirement for a segmentation that will be used in quarterly portfolio planning.

K. Project Outcomes

```
In [429... experiment_outcome = 'Hypothesis Confirmed'
```

```
In [430... # Do not modify this code
print_tile(size="h2", key='experiment_outcomes_explanations', value=exper
```

experiment_outcomes_explanations

Hypothesis Confirmed

```
In [431... experiment_results_explanations = """
Outcome: The clustering experiment confirmed that the 254 sales-active pr
separate into four structurally distinct commercial groups based on price
margin percentage, and discount behaviour. The hypothesis that meaningful
risk archetypes exist within the portfolio – and that they can be identif
from commercial feature data alone – is supported.

Key insights:
```

1. The dominant split in the portfolio is between low-cost high-margin products (Efficient Core, 95 products, 58.6% avg margin) and mid-tier products with compressed margins (Standard Mid-Tier, 108 products, 39.4% avg margin). Together these two groups account for approximately 80% of the portfolio and represent the baseline for pricing governance.
2. A premium group (Premium Discount-Exposed, 28 products) carries the highest dollar risk in the portfolio – not because its margin percentage is unacceptably thin (~40%, similar to Clusters 1 and 3), but because a 16.9% average discount on a \$2,783 average list price represents approximately \$470 of revenue given away per sale. These products warrant the most urgent pricing review.
3. The comparison between Clusters 2 and 3 is the most commercially significant finding: both are premium, high-cost product groups with approximately the same margin percentage. But Cluster 3 (Premium Price-Protected, 23 products) sustains its prices with just 1.5% average max discount, while Cluster 2 is exposed to 16.9%. Cluster 3 demonstrates that premium pricing is achievable for this product type – the problem in Cluster 2 is operational, not structural.
4. Two independent algorithms (KMeans and Mean Shift) both independently identified four density modes in the feature space, strengthening confidence that the four-archetype structure reflects genuine patterns rather than a modelling artefact.

Recommended next steps:

1. Pricing review for Cluster 2 (Premium Discount-Exposed): audit whether high discount rates applied to these 28 products are commercially justified or are eroding margin without equivalent volume uplift. At approximately \$470 of revenue given away per sale, even a modest reduction in discount exposure on these products has an outsized impact on portfolio margin.
2. Discount governance for Cluster 1 (Standard Mid-Tier): establish discount band limits for this group – margins are already below average and further discounting directly reduces profitability.
3. Price protection for Cluster 3 (Premium Price-Protected): maintain current pricing discipline. Cluster 3 serves as the benchmark – it proves that premium pricing works without heavy discounts for this product tier. A drift towards higher discount exposure risks replicating the Cluster 2 problem.
4. Refresh the clustering quarterly or after material changes to the product mix, cost structure, or promotional calendar to ensure the archetypes remain current.

""""

```
In [432... # Do not modify this code
print_tile(size="h2", key='experiment_results_explanations', value=experi
```

Outcome: The clustering experiment confirmed that the 254 sales-active products separate into four structurally distinct commercial groups based on price, cost, margin percentage, and discount behaviour. The hypothesis that meaningful margin-risk archetypes exist within the portfolio — and that they can be identified from commercial feature data alone — is supported. Key insights: 1. The dominant split in the portfolio is between low-cost high-margin products (Efficient Core, 95 products, 58.6% avg margin) and mid-tier products with compressed margins (Standard Mid-Tier, 108 products, 39.4% avg margin). Together these two groups account for approximately 80% of the portfolio and represent the baseline for pricing governance. 2. A premium group (Premium Discount-Exposed, 28 products) carries the highest dollar risk in the portfolio — not because its margin percentage is uniquely thin (~40%, similar to Clusters 1 and 3), but because a 16.9% average max discount on a

2, 783 average list price represents approximately 470 of revenue given away per sale. These products warrant the most urgent pricing review. 3. The comparison between Clusters 2 and 3 is the most commercially significant finding: both are premium, high-cost product groups with approximately 40% margin percentage. But Cluster 3 (Premium Price-Protected, 23 products) sustains its prices with just 1.5% average max discount, while Cluster 2 is exposed to 16.9%. Cluster 3 demonstrates that premium pricing is achievable for this product type — the problem in Cluster 2 is operational, not structural. 4. Two independent algorithms (KMeans and Mean Shift) both independently identified four density modes in the feature space, strengthening confidence that the four-archetype structure reflects genuine patterns rather than a modelling artefact. Recommended next steps: 1. Pricing review for Cluster 2 (Premium Discount-Exposed): audit whether the high discount rates applied to these 28

products are commercially justified or are eroding margin without equivalent volume uplift. At approximately \$470 of revenue given away per sale, even a modest reduction in discount exposure on these products has an outsized impact on portfolio margin.

2. Discount governance for Cluster 1 (Standard Mid-Tier): establish discount band limits for this group — margins are already below average and further discounting directly reduces profitability.

3. Price protection for Cluster 3 (Premium Price-Protected): maintain current pricing discipline. Cluster 3 serves as the benchmark — it proves that premium pricing works without heavy discounts for this product tier. Any drift towards higher discount exposure risks replicating the Cluster 2 problem.

4. Refresh the clustering quarterly or after material changes to the product mix, cost structure, or promotional calendar to ensure the archetypes remain current.